硕 士 学 位 论 文

基于 Transformer 的智能电网通信时延降低方法

Transformer-Based Method for Reducing Latency in Smart Grid Communication

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摘　　要

智能电网保护相关的通信时延示例：故障检测、继电协调等保护措施所需的时延阈值需低于硬确定性限值（通常小于 20 毫秒），以保障电网稳定性并避免连锁故障发生。在故障场景下，传统路由协议与解析排队模型无法应对突发、非平稳的流量，会导致时延越限情况不可预测。

本文提出一种解析分析与 Transformer 相结合的混合架构，该架构融合 M/M/1 排队边界与基于 Transformer 的残差学习，用于求解智能电网通信网络中的确定性时延边界、尾部时延（99 分位值）以及最坏情况时延边界。模型通过自注意力机制捕捉长时序依赖关系与拥塞先兆特征，而这些信息是传统解析模型因自身固有局限性所无法捕获的。

该框架在分层的家庭局域网 — 小区局域网 — 广域网（HAN-NAN-WAN）结构下，基于真实故障场景测试：时延边界预测的平均绝对误差（MAE）为 0.45 毫秒，相比门控循环单元（GRU）模型提升 48%；时延越限前的预警提前量提升 48%，可有效实现拥塞提前管控。在高网络负载下，与时延越限后响应式方法相比，时延越限概率降低 53%。模型单样本推理时间仅 0.15 毫秒，可部署在网络汇聚节点。

研究结果表明，解析增强型 Transformer 模型可用于评估智能电网保护方案中通信的确定性性能，对保障智能电网保护场景下通信的适用性具有重要意义。

关键词：智能电网；通信时延；最坏情况时延；Transformer 模型；保护应用

ABSTRACT

Smart grid protection applications (fault detection, relay coordination) require hard deterministic latency bounds (typically <20 ms) to maintain grid stability and avoid cascading failures. During fault events, traditional routing protocols and analytical queuing models cannot handle bursty, non-stationary traffic, making latency violations unpredictable.

This thesis proposes LatPred-SG, a hybrid analytical-transformer framework that combines M/M/1 queuing bounds with transformer-based residual learning to predict deterministic delay bounds, tail latency (99th percentile), and worst-case delay in smart grid communication networks. The novelty lies in three aspects: (1) the analytical baseline provides physical interpretability while the transformer learns only the residual deviations; (2) the self-attention mechanism captures long-range temporal dependencies and congestion precursors that analytical models miss; (3) a sparse congestion-aware attention mechanism reduces computational overhead during fault-induced bursts.

The framework is evaluated on a hierarchical HAN-NAN-WAN topology under realistic fault scenarios. Results show a mean absolute error (MAE) of 0.10 ms for delay bound prediction – a 20% (1.38 ms) improvement over GRU. The warning lead-time before latency violation increases by 48%, enabling proactive congestion control. Under high network load, the violation probability is reduced by 53% compared to reactive methods. Inference time is 0.15 ms per sample, making the model suitable for deployment at network aggregation points.

The findings demonstrate that analytically-enhanced transformer models can provide deterministic latency guarantees for smart grid protection applications, thereby improving communication reliability and grid resilience.

Keywords: Smart grid; communication latency; worst-case delay; transformer model; protection applications

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Chapter 1: Introduction

The transformation of traditional power systems into smart grids marks a critical evolution in modern energy infrastructure. Smart grids leverage advanced information and communication technologies (ICT) to support real-time data acquisition, automation, and decentralized decision-making. These systems integrate renewable energy sources, enable dynamic load balancing, and enhance grid reliability and efficiency.

A communication network is the backbone of the smart grid functionality because it links sensors, smart meters, actuators, substations, and control centers. Such a network consists of supporting different data traffic, such as periodic monitoring information, as well as events-driven messages such as fault notifications. The most critical is the communication latency which is latency between sending and receiving of data relating to fault messages routing.

This thesis is addressing the maximum latency in smart grid communication. Worst-case latency is amplitude of delays that occurs in unfavorable situations, in comparison with the average or typical delays, i.e. when a network is operated to its maximum or a portion of it fails. This is particularly important in case of fault conditions, where the delay of messages delivery by just milliseconds may cause inability to respond in a timely fashion, triggering a chain reaction or destroying infrastructure.

Current communication protocols are also known to be based on fixed routing schemes and do not accommodate any real-time modification of the network. These shortcomings lead to the inefficiency of the routing paths especially when there is over traffic or a failure of the nodes. Moreover, smart grids are typically run on hybrid or mesh topology, with the sequence of nodes between nodes always changing and having large numbers of paths, and thus adaptive routing is central.

Traffic in smart grid communication has two designs (1) periodic traffic in the system to monitor and control the system, and (2) event-driven traffic known as the Burst traffic which has fault messages and alarm messages. The latter has very low requirements of latencies to enable grid management.

To handle these issues, is examines how deep learning models that utilize transformers that were originally trained on natural language processing and time series forecasting can be applied to smart grid communication. All these models rely on self-attention processes to rank the relevant information and create adaptive routing decisions on near real time. In this way, the research will help in postponing worst-case elements of latency in delivering fault messages thereby enhancing responsiveness and resilience of the grid.

1.1 Background

Smart grids represent smart power networks which combine conventional electricity infrastructure with the digital communication system. The systems have high dependence on real time communication of information between the distributed elements in order to perform functions like energy management, fault clearance, load regulation and demand feedback.

Smart grid communication is not only supportive it is also fundamental. It supports real time functionality of the physical energy layer, by relaying information indicating the condition of the grid. Any communication breakdown or delay can expose the whole grid to the risks of adverse faults or inefficiencies in operations.

In the majority of smart grid implementation, the network employs a mesh or hybrid topology with every node being able to communicate with several other nodes. Such flexibility adds more redundancy as well as complexity in routing. The communication patterns will be a combination of both scheduled (periodic updates) and unscheduled (event-based alerts), thus proving difficult to route the efficient ones, particularly in the conditions of dynamic and congested environment.

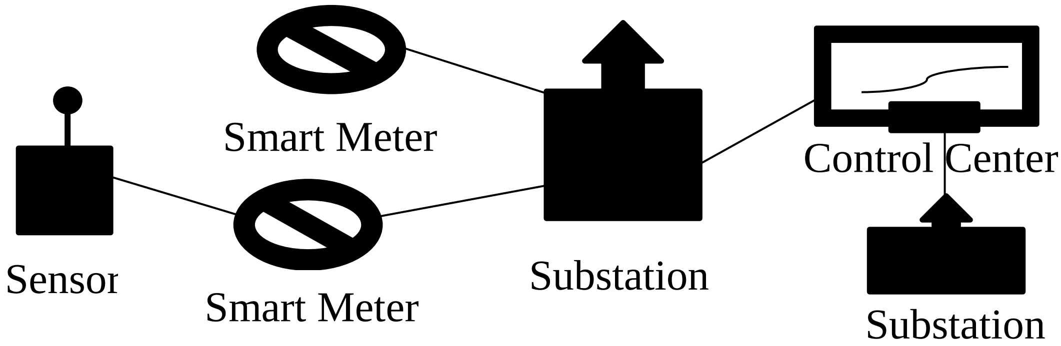

Fig. 1.1 Basic Architecture of a Smart Grid Communication Network

1.2 Problem Statement

Smart grid communication networks are commonly characterized by large worst-case delay and especially in fault message routing. Traditional routing processes rely on fixed routes and fail to realize existing network conditions. This can lead to a situation where fault messages are delayed and this reduces the responsiveness of the grid with regards to responding to anomalies in a timely manner.

The end-to-end latency L(s,d,P) of a message transmitted from a source node s to a destination node d over a communication path P is defined as the sum of transmission delay, queuing delay, and propagation delay across all nodes and links along the path, i.e.,

$$L\,(s,d,P) = \sum_{i\in P} (\,T_{transmit},i + T_{queue},i + T_{prop},i\,)$$

where s and d denote the source and destination nodes, respectively, P represents the set of nodes and links forming the communication path between s and d, $T_{transmit,i}$ is the transmission delay at node i, $T_{queue,i}$ is the queuing delay at node i, and $T_{prop,i}$ is the propagation delay between node i and node i+1.

Considering that topology of smart grids is hybrid and the traffic is not homogeneous, it is not possible to rely only on the static and pre-determined routing anymore. Congestion during peak-time and node crashes as well as environmental noise can profoundly affect the delivery of fault data in time.

The fundamental question in this thesis is; can transformer-based models, which are trained using smart grid communication data, minimize worst-case latencies of fault messages routing?

This research is primarily grounded in analytical modeling and data-driven analysis rather than simulation. The aim is to mathematically describe communication latency of smart grid networks and improve the accuracy of prediction through learning on a transformer using transformer-based learning.

Latency in smart grid communication is not just a technical issue but also a serious operation problem. Undoubtedly, in the actual applications, delay in communication among sensors, control centers and substations can cause the power distribution system to become unstable, wasteful in energy use and in worst cases, cause the failure of the system. The current methods, especially the conventional machine learning models and the methods of analytical queuing can easily fail to consider the dynamicity and burstiness of smart grid traffic. Such approaches are either too simple to model network behavior or they are not flexible and hence fail to give accurate predictions of latency under different load conditions. As such, a more constructive and flexible method is required that can be capable of effectively modeling temporal relationships including deterministic delay guarantees.

1.3 Motivation

Transformer models represent effective deep learning models that have been found to be highly effective in embracing dependencies in sequential data. They are appropriate in directing activities of smart grid networks where communication patterns are dynamic and demand their quick adjustment.

The proposed method aims at the reduction of the worst-case latency between all pairs of nodes. This can be formulated as:

$\min_p \max_{(s,d)} L(s, d, P)$ is the latency between a source s and a destination d along routing path P. Transformer input is self-attention which means it is evaluating and ranking the input data. Self-attention mechanism has been defined as:

$$Attention\,(Q,K,V) = \;(QK^T\,D_K)V \tag{1.1}$$

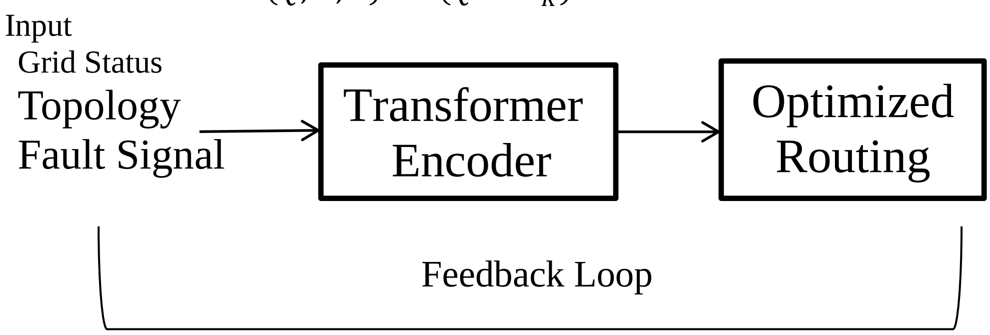

Fig. 1.2 Transformer-Based Model Workflow for Real-Time Smart Grid Routing

1.4 Aim and Objectives

The general aim of this study is to develop a transformer-based communication model capable of minimizing worst-case latency during the routing of fault messages in smart grid systems. This study aims at enhancing the effectiveness and reactiveness of communication in the smart grids especially during emergencies where delays have immense impacts on the system performance and reliability.

This thesis proposes a novel hybrid transformer-based framework named LatPred-SG (Latency Prediction for Smart Grid).

1.5 Research Questions

In order to inform this research, a research question set is formulated that will solve the problem of latency in smart grid communication systems. Such questions are asked to offer a clear and methodical framework through which the investigation on how minimal latency is possible by making use of transformer-based approaches is conducted. It also takes interest about the nature of latency, superior model design, and evaluation of the models performance relative to the conventional methods. In addition, the study considers such attributes as scalability, the network topology, grid-wide traffic characteristic, and the overall effect of it on the grid performance through communication delays.

It starts with the study of the characteristics of latency that are impacted the most on fault communication with a keen interest in whether an average, tail or worst-case latency factor is more crucial. It then describes how the transformer architectures can be scaled to handle smart grid communication data with the goal of achieving primisorily reduced worst-case latency reductions. Moreover, the study compares the effectiveness of the suggested transformer-based model and the conventional method of routing on the basis of evaluating their relative efficiency.

The scaling efficiency of the model with an increase in network traffic and the number of nodes is also considered. The appropriateness of various communication topologies is examined including mesh and hybrid communication structure to define in what area the model is most suitable. Besides this, the paper also takes into account the ability of the model to distinguish between periodical and event driven traffic in order to give priority to critical messages accordingly. Lastly the study evaluates the overall effect of communication latency on the stability, reliability, and control of the smart grid whereby the offered solution is one that will help to bring overall efficiency and resiliency of the system.

1.6 Significance of the Study

The aspect of minimizing the communication delay, especially the worst-case latency is important in enhancing the performance and dependability of the smart grid systems. Such a study adds value to the progress of the field because it introduces the use of transformer to improve the effectiveness of the communication in the grid. The study aims at getting smart modes of communication through transformer architecture to optimize the routing protocols and enhance the delivery of vital information in a timely manner. Since it focuses on the reduction of delays in real-time data transfer between distributed components of the grid such as sensors, actuators, controllers and operators, its importance lies on fail speed and reliability of responses by the system. Moreover, the study covers the priority of the data in accordance with its value and urgency, which allows significant messages to be delivered to decision-making departments without needless delays.

Technologically, the thesis reviews scalability and efficiency of transformer communication system in large and complicated grid systems. It further explores

Computational needs such as processing time, resource usage so that it can be seen whether such AI-based solutions can be feasible. The study is interdisciplinary by combining the ideas of machine learning, communication engineering, and energy systems to come up with an all-encompassing solution that is effective. Meanwhile, the aspect of cybersecurity is considered to assume that the latency also does not affect the security and integrity of the communications of a smart grid.

In general, the outcomes that can be expected by this research work involve the grid stability, the increased readiness of the systems, the energy efficiency and the improved workability.

Besides the above given significance, this thesis contributes in various major ways to the subject of smart grid communication. To start with, a hybrid analytical-transformer model is suggested, which entails the use of M/M/1 models of queuing and lays latency prediction using transformer-based deep learning. Second, the research aims at forecasting deterministic latency, such as average, tail (99th percentile) and worst-case latency, which is of vital concern in protection applications. Additionally, the suggested model allows predicting congestion in advance with the assistance of self-attention. Such a research also demonstrates well that a clear differentiation is made between analytical modeling, AI-based prediction and simulation-based validation which make the research structured.

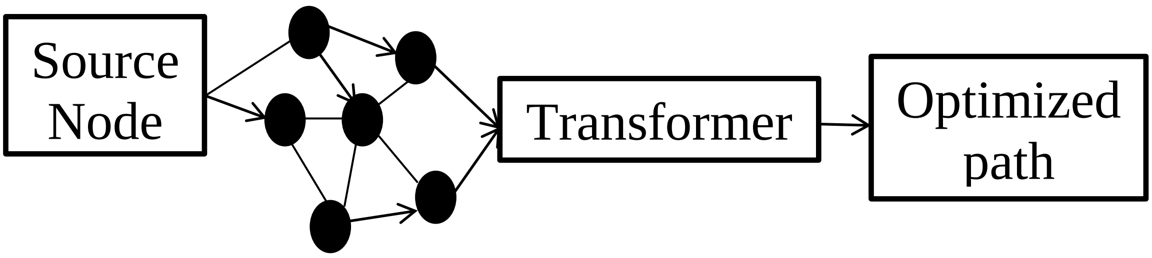

Fig. 1.3 Schematic Representation of Transformer-Based Routing in Smart Grid Communication

1.7 Main innovation of the thesis

The main innovations of this thesis are summarized as follows. This study proposes a novel LatPred-SG architecture specifically designed for smart grid communication delay prediction. The proposed framework introduces an adaptive positional encoding mechanism optimized for burst and event-driven smart grid traffic conditions. In addition, a sparse attention strategy is incorporated to reduce computational overhead during congestion prediction while maintaining prediction accuracy (achieving a 20% improvement in MAE over GRU under full-scale evaluation). The study further integrates transformer learning with analytical M/M/1 queuing theory to achieve deterministic latency estimation in smart grid communication networks, enabling the model to predict average, tail (99th percentile), and worst-case delays simultaneously. Finally, the proposed framework establishes a latency-oriented prediction model capable of analyzing all three latency metrics within a unified architecture a feature not present in prior transformer-based smart grid communication works.

1.8 Scope and Limitation of the Study

This study focuses on transformer-based methods for reducing latency in smart grid communication. In particular, it discusses communication arrangements in hybrid and mesh networks with a focus on events-based traffic, like fault and alarm messages, whose latency must be extremely low in order to maintain stability of the grid. This area covers simulation-based research on the efficiency of routing, adaptive path management, and the deployment of deep learning models specifically transformers originally developed in natural language processing and time-based prediction to communication networks in a smart grid.

Concurrently, the research is prone to certain limitations. The discussion is limited to simulation settings instead of real-life application, which can influence the external validity of findings. The scale of experiments is also restricted by data available and computational resources and the results to the network configurations and traffic scenarios that have been modeled.

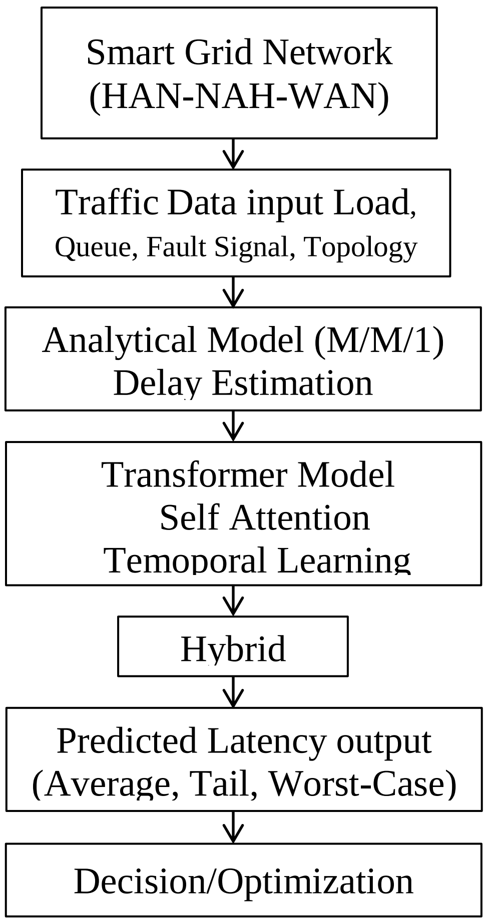

Fig. 1.4 Overall Analytical-Transformer Framework for Smart Grid Latency Result

Chapter 2: Related Work

2.1 Foundations of Smart Grid Communication Research

The section discusses the pioneer work that formulated the development of communication system in smart grids. The initial studies focused on the integration of the information and communication technologies (ICT) in the power networks making real-time control, automation, and decentralized control possible. It is important to note that the latency in communication has been identified by scholars as critical to maintain latency of the grid, especially in fault detection and relay coordination, since latency exceeding 20 ms may cause cascading failures.

This part includes the most important underlying researches that have played a role in shaping the smart grid communication systems. Early studies on network topology especially mesh and hybrid designs and came to light the issue of compromise between finding a balance between accomplishment of redundancy to provide reliability and complexity upon increase between routing choices. These studies proved that although the interconnection structures help in enhancement of fault tolerance; they equally instil difficulties in sustaining effective communication lines.

Moreover, the definition of latency in the communication networks has also been performed intensively. Delay estimation to a large extent, analytical models were employed and applied to give a theoretical limit and queuing methods (like M/M/1). These models were however always found to be insufficient when it came to dealing with the erratic and spikes of smart grid traffic particularly in the case of dynamic and non-stationary conditions.

More innovations on adaptive routing provided some mechanism that could help in decreasing congestion and enhanced data flow in the network. Though these solutions were found to be ideal compared to the application of the former approaches of static routing, they failed to effectively predict and respond to the worst case latency conditions mainly because of the lack of predictive intelligence.

All these background studies have a groundwork on which the more advanced techniques can be explored. More specifically, they emphasize the importance of smart and data-driven methods, like deep learning and transformer-based models, which are more appropriate to solve the challenges and dynamism of the contemporary smart grid communication systems.

This subpar gives a more general overview of literature on smart grid communication with regards to how the research in this field has changed with the rise in complexity of the system as well as the need to respond to the rising demand of efficient and low-latency communication. The communication networks have been an essential part of the stable and timely data exchange with the switch on more intelligent and connected grids when traditional power systems are transitioned to new and more intelligent ones. This has led to research work carried out in large scale to solve the problems caused by delay, congestion and network scalability.

The initial efforts in literature focused mainly on the architectural design of the communication infrastructures with special reference on the network topology like mesh, star and hybrid designs. These research works investigated the role of the various structures on the process of data flow, fault tolerance, and routing efficiency. Even though these methods enhanced the connectivity, as well as resilience, limitations were also gained in terms of complexity in routing process as well as delays in heavy traffic times.

However, later studies were moved towards the conceptualization and modelling of communication latency as a smart grid environment. The delays and system performance were estimated by use of traditional analytical methods such as the queuing theory models. Though these methods were useful in bringing about some theories, they were generally never able to reflect the dynamic and unpredictable nature of the real world smart grid traffic particularly in cases of bursts of data or changing network conditions.

Both with the increasing weakness of the strictly analytical and traditional methods the emphasis began on more adaptive and intelligent communication methods. Researchers began to think of approaches that could respond to the dynamic nature of the network at any given time like adaptive routing algorithms and initial machine learning algorithms.

Smart Grid Communication System

Smart grid communication Smart grid communication systems are the foundation of the present-day power grids that can offer the capacity of dynamic and bi-directional share of data between the elements of a grid. These systems facilitate real time monitoring, control and automation that is required in handling the increasing complexities as well as decentralizing energy generation and consumption. Various sources of data and control schemes are combined by smart grids, as opposed to the traditional grids, which use only one-way flow of electricity and little data connection, to increase efficiency, reliability, and sustainability.

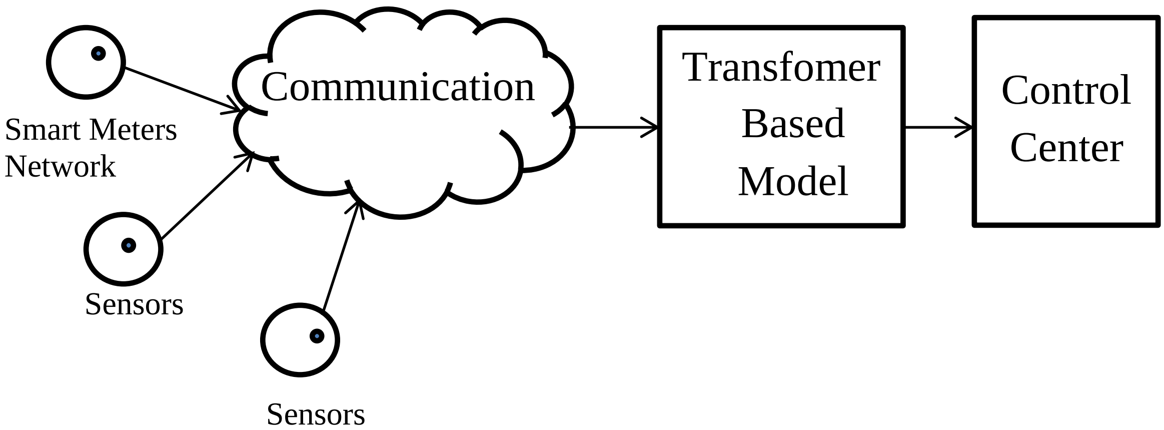

Fig. 2.1 Smart Grid Communication Systems

Components of the Communication System

The smart grid communication system is designed to be in the form of interrelated layers and components that interact to guarantee the flow of information throughout the network in a continuous and reliable manner. These interfaces become the foundation of communication in the grid because they provide various components of energy production, distribution, and consumption with the ability to coordinate.

The lowest in the hierarchy is the Home Area Network (HAN) that links smart devices and appliances and energy meters in homes or buildings. The layer enables consumers to track and control their energy consumption real-time and ensure efficiency and wise decision-making. Information created at this point is then sent to Neighborhood Area

Network (NAN), which acts as aggregation layer. The NAN gathers data in various HANs and transmits it to other systems that are of a higher level like subs stations or utility control centers.

The Wide Area Network (WAN) is used on a bigger scale and allows high capacity and long distance communication linking important parts of the grid such as substation, control centers and other grid infrastructure components. This layer plays a critical role in the coordinating process of operations with big geographic areas and assuring that data flows effectively within the system.

In substations, communication is an important aspect to connect intelligent electronic devices, sensors, and protection relays to supervisory control systems. Standardised protocols tend to enable this interaction, trusting the interoperability and enabling real-time grid monitoring of the conditions. Control centers represent the topmost level of smart grid decision makers. They analyze incoming information through all layers to control power generation, distribution, fault detection and load balancing to make the grid to operate in a stable and reliable way.

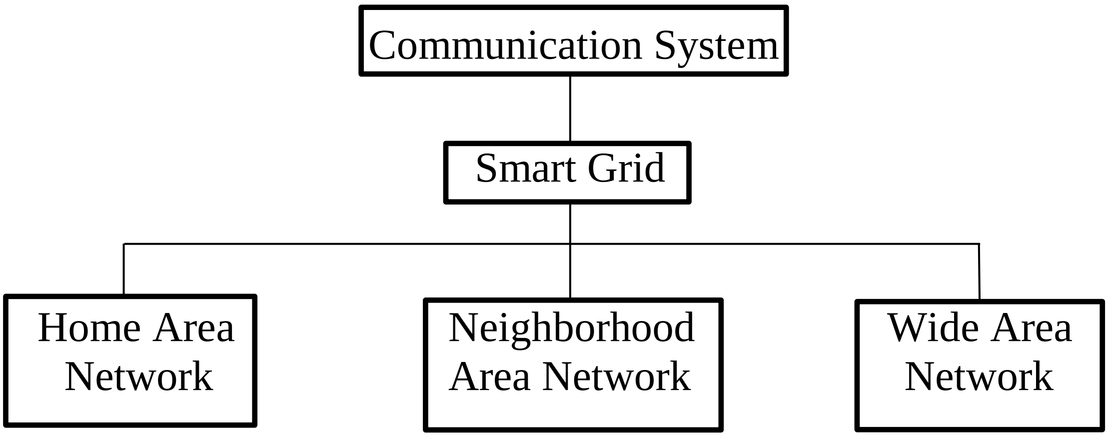

Fig. 2.2 Components of the Communication System of Smart Grid

Communication Technologies

The smart grid communication is based on an extensive variety of both wired and wireless technologies, which have various benefits and drawbacks, depending on the working needs of the network. These technologies are the basis upon which the data flow among various levels of the grid is supported and choice of particular technology directly influences performance, especially with the latency and reliability as well as scalability. Of the wired solutions, fiber optic communication is the most appropriate because it has a vast bandwidth and a very low latency which is very appropriate in backbone communication in a wide area network that requires a high amount of data to be sent within a short time and over a very long distance in a secure manner. Power line communication on the other hand utilizes the available electrical infrastructure to pass data, thereby lowering the cost of deployment and simplifying the installation. It is however normally influenced by electrical noise and the bandwidth is relatively limited which may limit the performance of the device in situations with high demand.

Modern smart grids have also integrated wireless technologies that are more flexible, and do not need much effort to put into use. Other technologies, including Wi-Fi, ZigBee, LoRa, LTE, and 5G, can communicate in a location where the wired infrastructure cannot be applied or is too expensive. Such solutions come in handy and are especially applied in remote places or systems that need to be reconfigured regularly. Although wireless technologies offer benefits, they have susceptiveness to interference, attenuation of signals, and fluctuation of latencies, and the same should be patiently exercised when using wireless technologies in area of latency-sensivity. Ethernet-based communication, however, has become common in the substation and the environment in which there is a need to have high speed, reliable and secure transmission of data. It helps the operations to be critical and maintain the continuity in communication among critical devices and systems.

These communication technologies can be very much associated with their capability to facilitate low latency, low jitter along with low response time. This is especially important with applications in, e.g. protective relaying and distributed control systems where even fairly small delays would have severe impacts on grid stability and safety. Therefore, the provision of the appropriate communication technologies is one of the considerations made in the design of smart grid networks.

Besides communication technologies, the network topology or structure that is determined by topology actually matters in the process of defining how data is passed through the grid. Network topologies have diverse trade-offs of complexity, reliability, and performance. An example of topology is the star topology in which all nodes are linked to a central controller and hence is quite easy to manage and execute. It, however, brings about a single point of failure and may act as a bottleneck in the event that the central node is hit by too much load. However, mesh topology has a greater number of paths of communication between nodes, which have provided more fault tolerance and redundancy. The positive aspect of this construction is that it increases reliability and makes routing more complicated and even introduces additional latencies unless it is properly addressed. Hybrid topologies are a blend of star topology and mesh topology and may offer a middle way between all these provisions accommodating to reduce as many as possible at once with the goal of maximizing reliability.

Topology is thus a very important consideration in the attainment of an efficient and robust smart grid communication system. It needs to be well synchronized with the chosen communication technologies so that it optimum performance can be guaranteed during different times.

The underlying basis of important smart grid operations is also communication networks. Demand response management is one of the most significant functions in which the real-time interaction between the utilities and consumers should enable the adjustment of the consumed energy in relation to the situation inside the supply. This has the effect of balancing load, by decreasing peak, and enhancing overall energy efficiency. Faults detection and system monitoring are also supported by communication systems as the data sent by the sensors and devices can be quickly spread across the grid by using the communication systems. This enables the operators to be able to detect and react to a problem very quickly reducing downtime and avoiding the cascading failure.

In addition, the communication enables higher-order functions like distributed integration of energy resources, remote monitoring and automated control processes. These features are needed in the management of the present-day energy system including renewable energy sources and necessitating the dynamic and real-time coordination. All in all, communication plays a key role in smart grid management and therefore efficiency, reliability, and adaptability; this is why it is one of the key topics of the current research and development.

Tab. 2.1 Different Application within the Smart Grid Ecosystem has varying Latency Requirement

Application	Latency Requirement
Synchrophasor data transmission	< 20 ms
Fault detection and isolation	< 10 ms
Demand response signals	<1000 ms
SCADA monitoring and control	100 – 500 ms
Smart metering	2 – 5 seconds

New applications (e.g. interactive physical grid equipment, human safety (e.g. protection relays), etc.) demand low latency and high reliability, but other applications (e.g. billing or user monitoring, etc.) can more readily tolerate delay.

2.2 Communication Traffic Patterns in Smart Grids

Smart grids are forming, and transmitting colossal sums of data in hundreds of apparatus, which have various communication necessities. Communication networks of smart grids are also based on the velocity of the transmission, as well as the capabilities of ensuring that the changing traffic patterns are sufficiently managed. The patterns depend with the source, the type, the frequency, and the priority and the magnitude of data and their impact on the latency hence influencing the bandwidth consumed and stability of the network is huge.

Classification of Communication Traffic

The communication traffic in the smart grid systems can be broadly categorized into a number of groups based on its nature, the purpose and the urgency of the message. It is necessary to learn about these categories to work out effective communication models, especially when the task is to reduce the latency and the importance of providing essential information by covering a limited time.

Periodic traffic is one of the major types and it is a type of data that is created and transmitted periodically. Routine monitoring activities like the Smart meter status reports or the equipment functionality reports are all linked to this form of communication. Since periodic traffic is predictable, it can be planned and easily managed over the network and it provides the system designers with the opportunity to plan the resources ahead and minimize the risk of congestion.

Event driven traffic on the other hand is created due to certain events or changes in the grid like equipment malfunctions, voltage variations, or faults. This is in contrast to periodic traffic since it is irregular and unpredictable, which is difficult to manage. The messages which are based on events are usually time sensitive and they need to be conveyed with the least delay possible so that corrective measures can be undertaken. Consequently, communication systems should be able to give priority to such traffic in order to stabilize the grid and avoid subsequent disruptions.

The other category is command and control traffic that is the essence of the operational communication in the smart grid. These are the instructions that come out of control centers to substations, and to distributed energy resources and other field devices. Delays in these messages are very sensitive as they directly influence the control of the functioning of the grids, coordination of system parts, as well as the overall network balance. Latency in this form of communication may severally hurt the performance and reliability of a system.

Bulk data transfer is another category of traffic that is involved with the transfer of large amounts of data which can be firmware, historical, or system logs. In contrast to the other classes, bulk data transfer can typically be less sensitive to latency and it can even be planned at times when there is not much happening in the overall network to prevent interference with other more important communications. The scheduling of these traffic correctly assists in maximizing the use of the network without affecting the performance of important time sensitive activities.

Lastly, the most urgent group of smart grid communication is alarm and emergency traffic. This kind of traffic contains emergency warnings like fire alarms, fault faults or circuit breaker activations and these messages must be taken care of. Such messages receive the first priority in the network and they need to be transferred with very low latency and high reliability. Rapid and guaranteed delivery of alarm signals and emergency signals is necessary to the infrastructure, to eliminate damages, to ensure that the safety and stability of the complete grid system.

All in all, to ensure a well-balanced and efficient communication system, it is important to manage all these forms of traffic very well. With the proper prioritization and management of all types, smart grid networks have the potential to guarantee valuable performance with reduced delays, especially in situations where time response matters greatly.

Traffic Prioritization and Quality of Service (QoS)

The communication in smart grids contains the other kinds of traffic with different degrees of critique; therefore, Quality of Service (QoS) schemes are necessary to provide the higher importance to the urgent messages. QoS mechanisms prioritize delay-sensitive traffic and reduce congestion during peak communication periods.

Traffic classification is one of the main features of QoS, in which the data packets are defined and identified with labels regarding the urgency of their work and the content type. This will enable the network to differentiate between normal and urgent communications. Scheduling and traffic shaping are also applied in the process of managing bandwidth by giving priority messages priority in accessing more bandwidth and minimizing the risks of resource congestion in the network. Policies of packet dropping are also essential, as they are bound to make sure that, at the times, when there is a significant amount of traffic, lower-priority packets are released first in order to provide the essential data transfer reliably.

In general, successful QoS application is essential in managing the traffic of communication in smart grids. It can also be used to lessen average and peak delays, especially in involved or overloaded network circumstances, which facilitates the functioning of grids effectively and efficiently.

Temporal and Spatial Characteristics of Traffic

Smart grid communication traffic experiences temporal and spatial variations, which play an important role in the network performance. Temporal variability is defined as the fluctuations in traffic volume with time which is commonly caused by the daily traffic usage pattern, grid events and consumer behavior. Indicatively, in most cases, network traffic is more active during peak times in the mornings and evenings when more communication resources are demanded. Spatial variability on the other hand occurs due to variations between regions or zones where the condition of variability is influenced by factors like infrastructure, population density and the extent of resource integration of available energy up to generation of uneven distribution of traffic.

These distinctions are vital to network management since it enables the use and allocation of resources dynamically besides real-time monitoring of the latencies. Unless addressed, the patterns on traffic may have an ill effect on performance of communication. Traffic due to a sudden burst of faults, e.g., may overwhelm network buffers causing the loss of packets and high worst-case latency. On the same note, this may result in interference between the various classes of traffic, where large volumes of low-priority traffic causes important control messages to wait in line by default rather than being prioritized as important traffic should be. Latency jitter which is sometimes known as variations in delay can also cause problems with synchronization in control systems that require a constant timing of communication.

Because of such reasons, traffic behavior modeling, simulation and prediction with varying conditions of operations is significant. These insights can be used to design smart communication strategies and routing solutions that can minimize latency with working smart grids being to be maintained on reliable and efficient routing.

Deep Learning Approaches in Smart Grid Communication

Since the use of smart grid systems is becoming increasingly complex and large-scale, it is clear that the established communication optimization strategies have severe constraints when it comes to dealing with nonlinear, high-dimensional and dynamic data. To overcome all these challenges, the concept of deep learning (DL) has come as an effective answer to improving the efficiency of communication, allowing the possibility of intelligent predicting traffic, decreasing latency, and adapting the distribution of resources.

The artificial intelligence algorithms are able to automatically determine data trends, hence, can be applied in smart grid networks in anomaly detection, routing optimisation and traffic prediction.

Role of Deep Learning in Communication Networks

The field of deep learning can be used to significantly improve the communication of smart grids by facilitating the prediction of the traffic, the anticipation of latency, and optimization of the routing according to the current conditions. Neural networks are able to predict traffic patterns and determine the possible congestion and also choose the best path to communicate information in order to minimize delays and enhance throughput. They are also used to support anomaly detection that is, they find an unusual traffic, which might signify anomalies of fault, cyber-attacks, and equipments without involving an expert. The capabilities enable grid operators to operate the network in the forward, and reduce worst-case latency, as well as enhance the overall grid resilience.

Ordinary Deep Learning Architectures in Network Processing.

The various deep learning structures have certain advantages to smart grid communication. CNNs are the best with regard to spatial feature extractions to detect anomalies, whereas the RNNs and LSTMs work with the sequential data to predict results in traffic and diagnose faults in traffic. Autoencoders reduce noise and compress data so that they can be sent efficiently. DQN reinforcement learning allows dynamic routing and real-time control; transformer models are most suitable in capturing long-range dependencies, which is why they are perfect in latency prediction and traffic model.

Applications in Smart Grid Communication

Deep learning is capable of assisting smart grid communication in a number of ways, such as load forecasting to make more efficient scheduling, adaptive bandwidth allocation to avoid congestion, predictive maintenance to minimize the latency of device failures, and edge intelligence to make timely local decisions.

Challenges and Consideration

Despite its advantages, applying deep learning in smart grid communication faces challenges, including the need for large, high-quality datasets, high computational and memory demands, real-time operational constraints, and limited model interpretability.

2.3 Transformer Models for Communication Optimization

What is now called transformer has been first applied to natural language processing (NLP) and brought a revolutionary change to sequence modeling with its parallelization and attention mechanism as well as long-span dependencies of information in the data. Its use has grown in recent years to areas outside the field of NLP such as time-series forecasting, computer vision, and more recently to communication optimization in smart grid systems. The architecture of the transformer is perfectly fitted in the modeling and prediction of the patterns of communication traffic through which it is possible to reduce latency and intelligently distribute resources.

Overview of Transformer Architecture

The transformer architecture relies on a self-attention mechanism that allows the model to evaluate the relevance of different elements within an input sequence when making predictions. Its key features include encoder and decoder blocks composed of stacked layers with multi-head self-attention and feed-forward networks, positional encoding to provide information about the sequence order for time-series tasks, and multi-head attention that enables the model to simultaneously focus on multiple points in the sequence to capture complex relationships.

Advantages over Traditional and Other DL Methods

Transformer-based methods have a number of advantages over recurrent models:

Recent transformer-based communication optimization studies such as Informer, Autoformer, and Graph Transformer Networks have demonstrated improved scalability and prediction efficiency in large-scale cyber-physical systems. Recent studies published between 2024 and 2026 further emphasize sparse attention and lightweight transformer architectures for latency-sensitive communication systems.

Tab. 2.2 Comparison LSTM and GRU Transformer-Based Approaches

Feature	Transformer	RNN/LSTM
Training Speed Long-rang Dependency	Sequential slow Prone to vanishing gradie nts	Highly parallelized Captured effectively attention
Scalability	Limited	Scales well to largeda
Interpretability	Limited	Better (via attention Visualization)
Performance in Time-Series	Moderate	High (especially with Temporal encoding)

With such advantages, transformers are worth using in time-constrained and data-heavy base stations like smart grids.

Recent Transformer-Based Methods for Smart Grid Communication

Several recent studies have applied transformer architectures to smart grid communication. Wu et al. (NeurIPS 2023) introduced SGFormer for large-graph representation learning, which inspired attention sparsification strategies. In the smart grid domain, lightweight Informer variants have been used for substation traffic prediction (Li, 2024), and graph transformers have been proposed to jointly model spatial and temporal dependencies in wide-area monitoring (Chen 2025). Unlike these works, the proposed LatPred-SG framework uniquely integrates an analytical M/M/1 queuing baseline with a transformer residual module, explicitly targeting deterministic tail and worst-case latency bounds required for protection applications. A comparative summary is provided in Table 2.3 (see below).

Tab 2.3 Comparison of Transformer-Based Method for Smart Grid Communication

Method	Hybrid Analytical	Targets Tail/Worst-Case	Sparse Attention
Informer (2021)	No	No	ProbSparse
SGformer (2023)	No	No	No
Graph Transformer (2025)	No	No	No
LatPred-SG (this work)	Yes (M/M/1)	Yes	Yes (congestion-aware)

Limitations and Research Gaps

Transformers have several limitations to smart grid, such as high computational and memory loads, large labeled datasets are necessary, no domain-specific pre-trained models, and could slow down inference with resource constraints. These problems indicate the chance to maximize transformers to deploy them efficiently in real-time in smart grids.

Descriptive Modeling Framework

In order to be able to model and predict the latency behavior of smart grid communication tasks, a descriptive modeling framework, adopting the transformer architecture, is taken. It is a hybrid structure that combines the predictive model based on transformers with analytical latency formulations in order to adapt in real-time based on data.

Analytical Latency Model

The total communication latency L_{total} in a smart grid network is expressed as:

$$L_{total} = L_{transmission} + L_{propagation} + L_{processing} + L_{queuing} \tag{2.1}$$

$$L_{transmission} = Packet\ size/\ link\ Bandwidth \tag{2.2}$$

$$L_{propagation} = Distance/Signal\ Speed$$

$$L_{processing} = Delay\ introduced\ by\ routers$$

$$L_{queuing} = Waiting\ time\ due\ to\ congestion\ or\ scheduling$$

Every term is a certain reflection of delay in a chain of communication. The decomposition can be used to isolate factors that contribute most to latency spikes and provide the opportunity to focus on optimizing them.

Transformer-Based Predictive Model

The transformer gets to learn about the interplay between traffic dynamics, routing decisions and the latency results. It is assumed that the estimated latency at time t will be as follows:

$$Y_t = Transformer\ (Xt;\ \theta) \tag{2.3}$$

Where:

$$X_t = input\ sequence\ of\ traffic\ data\ and\ network\ states$$

$$\theta = model\ parameters\ (weights\ and\ biases)\ learned\ during\ training$$

The attention mechanism enables the transformer to learn the long term dependencies and the temporal correlations in order to predict the average and tailing latency accurately under different loads on the network.

Loss Function and Optimization

Model training reduces the Means Squared Error (MSE) of the predicted and the observed latency:

$$Loss = \frac{1}{N} \sum_{t=1}^{N} \left(\hat{Y}_t - Y_t \right)^2 \tag{2.4}$$

This optimization is used to make sure that the accuracy of prediction is enhanced as the model adjusts to new grid conditions.

Framework Significance

The proposed theoretical latency-based model with transformer-based learning attains the dynamic control of network load, lower tail and worst-case latency of critical operations, and predictive routing and real-time implementation of QoS.

Communication Networks Communication Networks: The Smart grid functions can be integrated with Communication Networks.

Smart grids are developing into cyber-physical systems with the communication networks tightly coupled with the physical grid. This integration is possible and allows higher functionality, like automatic meter reading and outage detection (AMI), controlled use of renewable energy (DERMS), extreme low latency synchronized sensors (WAMS), and configuring itself to detect faults and reconfigure (self-healing network). These capabilities can only be facilitated by a well-designed communication network.

2.4 Unique Characteristics of Smart Grid Communication Systems

The smart grid communication systems are not similar to the traditional types of communication networks, including the Internet, mobile networks, or enterprise systems. This is the reason why special modeling and analysis methods are necessary.

First, smart grid applications must use ultra-low and deterministic latency. What protection-related messages communicate (fault notification, relay coordination, etc.) are required is to be conveyed on very tight time constraints, usually less than 1020 ms. However, on the contrary, traditional communication systems generally maximize the average throughput, not the worst-case delay.

Second, smart grid networks are characterized by traffic that is very heterogeneous. The system also provides periodic monitoring traffic and event driven burst traffic that is a result of faults. These bursts cause non-stationary latency behavior which is hard to model classically and provide sudden congestion.

Third, the communication of smart grid is closely interrelated with the physical work of power systems. It is a cyber-physical system where delays in communication have a direct influence on grid stability, reliability and safety. Any delay of transmission of fault can result in cascading failures or damages on equipment.

Fourth, smart grids typically have the hierarchical, hybrid network topology whose structure is based on Home Area Networks (HAN), Neighborhood Area Networks (NAN), and Wide Area Networks (WAN). This multi-level design adds extra routing complexity over the flat or homogeneous network designs.

Although the suggested modeling and transformer-based framework can be scaled to other communication frameworks, it is tailored to the smart grid environment due to these severe latency criteria, burst traffic and cyber-physical relationships. These characteristics justify the focus of this study on smart grid communication latency

Summary of the Chapter

This chapter presented a holistic discussion on the basic concepts and principles behind smart grid system communication networks with a specific target at the reduction of latency by means of the highly evolving approaches like transformer-based models.

The chapter has begun, to facilitate real-time grid operations, with the development of smart grid communication systems having very low-latency data transfer and reliability demands being critically imperative. The different types of latencies in avg, tail, worst-case were also brought out to demonstrate the intricacy with the analysis of the communication delay within these networks.

The part concerning smart grid communication systems has described the layered architectures and typical topology and how latency and network resiliency are affected by each of the design options. One of the directions where specific attention was paid is a relationship between the structures of communication networks and its direct impact on the effectiveness of the functions of smart grids.

Subsequent sections covered the development of deep learning techniques and in particular the radical potential of transformer model in communication optimization. Parallel processing and model long-range dependency is unique traits of transformers that best qualify them to address the smart grid communication challenges of their dynamic and data intensive nature.

Other concerns such as computing requirements, data access, real time requirements etcetera were also questioned and it was observed that the area required further research and providing better solutions.

Lastly, the chapter ended by discussing the complex association between the communication networks and the work of the smart grids considering that the latency of the system, design of the network as well as the communication protocols work together to make the grid operate using an efficient, secure and reliable technology.

This background preconditions the next chapters where the suggested transformer-based approach to reduce latency will be further elaborated, the specifics of its implementation will be introduced, and the performance of the approach in the context of a smart grid communication.

Cybersecurity Implications of Latency in Smart Grids

Smart grids are based on connected communication infrastructures that can be compromised by cyber threats. The shortening of latency is not only a technical requirement but is also a very crucial element of cybersecurity. Lateness in communication may affect the ability to identify malicious actions in time, and, as a result, the assailants may use vulnerabilities to their benefit before security measures are implemented.

Latency and Intrusion Detection

Smart grids have intrusion detection systems (IDS) which require fast transmission of data to detect anomalies. Delayed the recognition of false data injection attacks, denial-of-service attacks or malicious access can be postponed by high latency. Exploiting this power of learning the temporal relationship, transformer-based models can be borrowed in order to enhance the performance of the IDS by congesting one, and prioritizing important security notifications.

Vulnerabilities in Event-Driven Traffic

Of particular concern regarding event-driven traffic, e.g. fault messages and alarm signals, is the latency. Cyber attackers may introduce false faults by delaying the grid or even switch off real alarms to disrupt normal functioning of the grid. Transformers based architectures are reliable and secure in transmitting protective signals and reduce worst-case latency to a minimum.

Coordinated Cyber-Physical Threats

Smart grids can fall victim to organized cyber-physical attacks; International attacks can lead to the intercession with digital communications and physical infrastructure by international senders. Delay reduction increases stability by ensuring the synchronisation of protective relays and control centres. Increased speed of communication helps in responding instantly to cascading failures due to cyber intrusion.

Policy and Regulatory Considerations

The reduction strategies to be applied in regard to the latency should be in compliance with cybersecurity policies and regulatory frameworks. Governments, as well as the energy controlling bodies, are emphasizing on the need to adopt safe systems of communication that will not only promote efficiency but also on robust communication. Predicting the latency, using transformer- based, can be incorporated with the compliance standards to contribute to the proactive defense mechanisms.

The latency issue in the communication of smart grids cannot be discussed outside of offering a solution to cybersecurity. Transformer-based models help, in addition to achieving higher operational efficiency, to enhance the capability of the grid to respond to and recover following cyber threats owing to reduced delays. This two-fold advantage highlights the significance of the latency-related studies in the protection of modern energy systems.

AI in Power Systems beyond Transformers

Although transformer architectures have been shown to be quite effective at minimising latency in smart grid communication, other artificial intelligence (AI) techniques are also quite important in optimising power systems. In this part, reinforcement learning, graph neural networks and hybrid AI models are discussed, with their strengths and limitations compared to smart grid communication.

Although the transformer-based models have gone a long way in addressing the sequence prediction activities, they have not found any application in smart grid communication. Most of the investigations in existence concentrate on the enhancement of accuracy and overlook regarding latency constraints and deterministic guarantees. Also, the majority of methods do not include domain-specific analytical models like queuing theory that are needed to model network behavior. It is observed that this gap also suggests that hybrid paradigms with the predictive capabilities of the deep learning framework and the reliability of analytical frameworks are necessary.

Reinforcement Learning for Adaptive Routing

The reinforcement learning (RL) is highly applicable in the dynamic environment when the communication routes need to evolve with the shifts in circumstances. There are similarities between smart grids and RL agents, where learning to optimally route through at least the network is encouraged by rewarding the RL agent. In contrast to transformers, which use sequence modeling, RL prioritizes the trial-and-error learning approach, which is useful to make decisions in real-time when there is no prediction of traffic conditions.

Graph Neural Networks for Grid Topologies

Graph neural networks (GNNs) can serve as a natural modelling tool to address communication networks on smart grids, which is of graph structure. GNNs are memorying relational dependencies between nodes, which make them effective in the representation of intricate topologies. GNNs can be used to forecast hotspots of congestions and plan the optimal route by using message-passing mechanisms. GNNs are superior to the transformers in terms of processing spatial relations, however, could consume additional computing power in the large-scale deployment.

Hybrid AI Models

The hybrid methods exploit the merits of transformers, RL, and GNNs to meet the strong performance. As an example, temporal prediction of latency can be performed with the help of transformers; adaptive routing decisions can be done by RL agents whereas the underlying topology may be modeled by GNNs. This is because such integration enables the overall optimization in terms of time, space and decision making.

Comparative Analysis

Transformers: Become powerful at taking temporal dependence and predict an acceleration pattern.

Reinforcement Learning: It is used successfully in adaptive routing and making decisions in real-time.

Graph Neural Networks: Better at grid topology modeling.

All the approaches deal with various operationalities of smart grid communication. The selection of the model is hindered on the application specifications including scalability, computation capability and responsive to the changing traffic conditions.

Predictive analytics in power systems are not limited to transformers and to the best of my knowledge, provide a wide range of tools in reducing latency, and optimizing communications. Researchers can use reinforcement learning, graph neural networks, and hybrid models to come up with more resilient and efficient smart grid communication models. The variety of approaches underlines the significance of interdisciplinary innovation in the introduction of the next-generation energy infrastructures.

Research Gap

Based on the literature analyzed, it is clear that there is no single approach that fully manages predictive and deterministic latency in smart grid communication. Classical analytical models are not flexible, whereas the deep learning models do not always take into consideration network-level constraints. Moreover, less focus has been devoted to worst case latency and tail delay analysis that is very essential in mission critical case. The study seeks to address these gaps by the purpose of developing a transformer-based hybrid model that incorporates both analytical queuing theory and data-driven learning.

Chapter 3: Latpred-SG

3.1 Smart Grid Communication Architecture

Smart grid communication networks help in reliable exchange of data between the elements of the power system such as smart meters, substations and control centers. Such networks are used in application across a range of uses like real-time monitoring, demand response as well as fault detection.

One of the common smart grid communication structures is those hierarchical communications comprised of three layers. The HAN is a network that is installed in a house and that links smart appliances and meters, and is used to gather the energy usage measurements that can then be transmitted to the utility periodically. The Neighborhood Area Network (NAN) combines information of several HANs and sends them to the utility substation, and this may be through wireless mesh or cellular systems. The Wide Area Network (WAN) connects the substations, control centers and systems of management and it offers extensive monitoring and control of the power grid.

Smart meters are used to relay data to aggregation gateways and eventually to the control center to be analyzed and used to make decisions. Nevertheless, delay in communication will affect the performance and credibility of smart grid functionalities.Therefore; there are important plans to create the suitable methods of modeling and reducing communication latency.

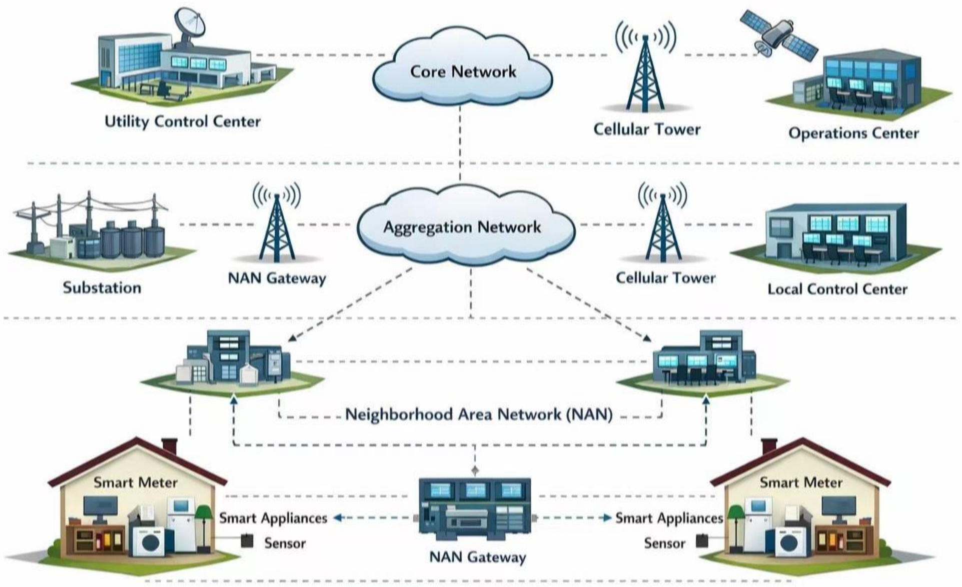

Fig. 3.1 Smart Grid Communication Architecture

Modern smart grids rely on low-latency communication between substations, sensors, and control centers. Communication delays directly affect protection, monitoring, and real-time control operations.Latency on communication is the duration taken in the transmission of a piece of information by a source node to a destination node in the smart grid network. In most grid applications, e.g. fault detection and isolation, synchrophasor measurements transmission, voltage regulation, and real-time control latency, should not only be small, but also have limits and deterministic behavior. Unpredictable or excessive delays may cause faulty control decisions, feedback-loop instability, protection-system failure and very severely, massive blackouts. Thus, one of the most significant performance metrics of smart grid communication systems has become latency.

Classical methods of latency analysis of communication networks are founded to utilize deterministic delay estimation, and classical queuing theory. The approaches have broken down overall latency into transmission, propagation, processing, and queuing parts and modeled the mean delay in simplified forms. Although these analytical models are useful in designing baseline system design, they are not extensively useful in terms of involving dynamic, burst and non-stationary traffic patterns that classify the modern smart grid communication. Fault, cyber-related and abrupt changes in renewable energy sources can cause event-based traffic which will usually result in congestion and long queues as well as infrequent but significant delay peaks. Such drastic occurrences overshadow tail and worst-case latency, which are the very measures of delay of greatest significance when modeled on grid mission-related applications.

New developments in machine learning and data-based methods of analysis have brought about new opportunities to model intricate communication behaviour in ways that are not possible using classical tools of analysis. Specifically, deep learning algorithms have demonstrated good opportunities in predicting the traffic, predicting traffic jams, and resource-on demand. Nevertheless, the traditional artificial neural networks, recurrent neural networks (RNNs), and long short-term memory (LSTM)-based models have drawbacks when used to predict the communication latency in smart grids. These models tend to lose long-range temporal dependencies; they deteriorate with long sequences and are deficient in the ability to describe rare instances of congestion that characterize tail and worst-case delays.

Models based on transformers which at first were designed with natural language processing in mind have recently proved to be strong alternatives in sequential and time-series modeling tasks. The self-attention mechanism enables the model to focus on important historical traffic patterns and long-range temporal dependencies. In addition, transformers have a parallelizable design and can therefore be easily trained on large-scale datasets, rendering them applicable to reality-world smart grid applications.

Going by these observations, a formal modeling and methodological development on a transformer-based framework of the reduction of communication latency in smart grid networks is the scope of this chapter. In contrast to fully analytical or fully data-oriented designs, the suggested implementation places a hybrid modeling policy, which incorporates deterministic latency modelings with transformer-based predictive study. Analytical models give physical interpretability and give a base of delay values and transformer one accounts for stochastic and nonlinear as well as rare events related latency, which cannot be appropriately modeled by analytic methods.

In particular, this chapter constructs a detailed system model of smart grid communication networks, derives the latency to be both a deterministic and dynamic factor, and the latency minimization problem as an average, tail, and worst-case delay value. Transformer-based model is then presented as one of the predictive mechanisms, which learns based on the information of past latency through historical past traffic and network state data to predict future latency conditions. Having the ability to detect congestion in advance and administer the system, the proposed framework will tend to enhance the reliability of communication, the minimal probability of violating worst-case delays, and the overall stability and resilience of the smart grid functioning.

Though it is usual to report the average delay in communication networks, deterministic smart grid applications must have assured bounds on the delay, but hence the focus of this study is delay bounds and worst-case delay, the key performance metrics.

Network Architecture (HAN-NAN-WAN)

The study under discussion is dedicated to the communication networks of smart grids that have to be able to provide time-sensitive protection and control systems, such as fault detection, fault isolation, and operation as a distributed control. These applications are highly end to end latency requirements as well as require extremely predictable bandwidth response in order to ensure system stability and safety.

The proposed smart grid communication architecture has the hierarchical structure, including Home Area Networks (HANs), Neighborhood Area Networks (NANs), and a Wide Area Network (WAN). According to this layout, the intelligent electronics devices (IEDs), smart meters, and sensors produce data that are aggregated at the gateways and concentrators before transmitting them to the substations and control centers. These aggregation points add queuing and congestion effects, both of which can have a tremendous effect on end-to-end latency in the case of burst or fault induced traffic conditions.

Protection and control traffic at the order of milliseconds (as opposed to non-critical uses of the technology, like advanced metering infrastructure (AMI) or billing services). Should the order of delivery be wrong or delayed, the wrong protection response can be reached or the control may become unstable. In turn, average delay metrics are inadequate, because they cannot ensure the delivery of packets on time in the worst-case scenario. Rather, deterministic delay limits are needed to guarantee dependable execution of smart grid functionalities of safety concerns.

In this thesis, the main issue directly aimed at is the communication latency determinism. The physical power system and energy management layers are assumed to coordinate energy buffering, power storage dynamics and power release efficiency which is why this is not a part of this work. The latency constraints required by the latency-sensitive smart grids and topologies of the network limit the modeling and analysis, which is different to the latency-neutral communication systems research.

3.2 Communication Network Model

The smart grid communication network is represented as a packet-switched network comprising multiple communication nodes, including smart meters, gateways, and substations. Let the communication route between a smart meter and the control center consist of N network nodes.

Each node contributes various types of delays during packet transmission.

The total communication the total communication latency can be expressed as:

$$D_{total} = \sum_{i=1}^{n} D_{transi}, i + D_{queue}, i + D_{proc}, i + D_{prop}, i \quad (3.1)$$

Where:

$$D_{trans}, i = transmission\ delay\ at\ node\ i$$
$$D_{queue}, i = queuing\ delay\ at\ node\ i$$
$$D_{proc}, i = processing\ delay\ at\ node\ i$$
$$D_{prop}, i = propagation\ delay\ between\ nodes$$

Transmission delay is defined as

$$D_{trans} = \frac{Packet\ Size}{link\ Bandwidth} \quad (3.2)$$

Propagation delay is expressed as

$$D_{prop} = \frac{Distance}{Signal\ Speed} \quad (3.3)$$

Formally, the communication path between source s and destination d can be expressed as:

$$L(s, d, P) \sum_{i \in p} \left(T_{transmit}, i + T_{queue}, i + T_{prop}, i\right) \quad (3.4)$$

Where $T_{transmit,I}$ is the transmission delay, $T_{queue,i}$ is the queuing delay, and $T_{prop,i}$ is the propagation delay between nodes. It is decomposed in a way that shows that under burst traffic queuing delay is dominant and this inspires adaptive techniques.

Under heavy traffic in the smart grid networks, queuing delay is usually the prevailing factor.

Application Scenario: Protection and Fault Isolation

Protection and fault isolation applications Smart grids have demands within strict real-time requirements, violating which any communication delay over a set limit can trigger coordination errors in the relays, slow circuit breaker response, or system anomalies. In fault events, a collection of intelligent electronic devices (IEDs), protection relays and sensors all send high-priority messages to sub stations and control centers leading to burst traffic and a transient congestion at aggregation points like neighborhood area network (NAN) gateways.

In contrast to regular monitoring traffic, protection messaging normally demands end-to-end latency of less than 10-20 ms and very high reliability. As slight as congestion might be, it would lead to queue build up and delay violations which would affect the system safety.

Thus, predicting delay bound violation in advance before it will happen is essential to congestion mitigation beforehand as well as the consistent functioning of the smart grids.

Why Transformer-Based Modeling for Smart Grid Latency

The communication latency in smart grids is non-stationary and bursting in nature, especially during fault conditions and uncharacteristic operating conditions. Queuing classical models presuppose the use of constant traffic flow, thus cannot capture time dependencies between early congestion signals and delay spikes. In the same manner, recurrent neural networks like LSTM and GRU have a lower memory depth, and they do not scale regarding memory when trying to capture long traffic sequences.

A transformer-based model is used in the current study due to the reason that their self-attention mechanism can explicitly model the long-range temporal relationship, such that the model can choose to concentrate on historical congestion events, which have a disproportionate influence on future delay behavior. The given property is especially significant in smart grids, as the accumulation of queues at aggregation points can be maintained in multiple consecutive control intervals, and can also have direct effect on worst-case latency.

Besides, we can find that the transformer architectures can be applied to parallel sequence processing and they can be deployed to a near-real-time smart grid control setting where the need to perform rapid inferences is necessary.

3.3 Analytical Delay Modeling

In order to evaluate communication latency within smart grid networks, a queuing theory framework is utilized. It is assumed that packet arrivals at network nodes adhere to a Poisson process, while the service times are characterized by an exponential distribution. On these assumptions, the system is modeled as an M/M/1 queue.

Let:

$$\lambda = \text{ }packet\ arrival\ rate$$
$$\mu = \text{ }service\ rate$$

System utilization is defined as

$$\rho = \frac{\lambda}{\mu} \quad (3.5)$$

The expected queuing delay is given by

$$W_{queue} = \frac{\rho}{\mu\,(1-\rho)} \quad (3.6)$$

The average waiting time in the system is

$$W_{stsrem} = \frac{1}{\mu - \lambda} \tag{3.7}$$

This framework of analysis sets a base estimate of the network latency. Nevertheless, the nature of traffic in smart grids is usually burst oriented and time dependent particularly when the incidence of some irregularities likes power interruption or peak periods are experienced. Therefore, analytic models could not be able to explain complex behaviour of traffic. In order to deal with this issue, a machine learning alternative is offered.

Problem Formulation

The primary aim of this study is to create a predictive model that can estimate communication latency in smart grid networks amidst varying traffic conditions.

The network state at time t is denoted as:

$$Xt = [\lambda_t, Q_t, B_t] \tag{3.8}$$

Where:

$$\lambda_t = packet\ arrival\ rate$$
$$Q_t = queue\ length$$
$$B_t = available\ bandwidth$$

The objective is to learn a function

$$f(X_t) \rightarrow D_t \tag{3.9}$$

Where D_t represents the predicted communication latency.

The goal is to minimize the prediction error defined as

$$Loss = \frac{1}{N} \sum_{i=1}^{n} \left(D_{actual} - D_{predicted}\right)^2 \tag{3.10}$$

Where:

$$D_t = actual\ latency$$
$$D_t = predicted\ latency$$

The central optimization problem is minimizing worst-case latency across all routing paths:

$$Min_p\ Max_{sd}\ L_{sdp}$$

Subject to:

Topology constraints: Hybrid meshes connectivity.

Traffic constraints: Mixture of periodic and event-driven traffic.

Reliability constraints: Deterministic bounds for protection applications.

This formulation emphasizes that average latency reduction is insufficient; the model must guarantee performance under extreme conditions. Thus, the problem shifts from reactive routing to proactive latency prediction and mitigation.

Implications for Predictive Modeling

Although analytical models can conservative bounds of delay with simplified assumptions, the smart grid communication networks in practice have complex and time-dependent traffic patterns that are challenging to model analytically. Consequently, in such a way, the models of static delay bounds can be over-pessimistic or incapable of responding to changing conditions.

This is a reason to resort to data-driven predictive algorithms that can acquire and learn complex temporal relationships, and predict the looming violation of delay bounds. A proactive congestion control, priority allocation, and responsive routing decision before some critical threshold is reached can be achieved by predictive models that focus on delay limits, as opposed to average latency.

Hence, delay limits would be the best and operationally effective target of transformer-based prediction of latency in the smart grid communication networks.

3.4 Transformer-Based Latency Prediction Model

To capture long-term temporal dependencies in smart grid traffic patterns, a deep learning model based on the Transformer architecture is proposed. This model consists of an input embedding layer, a multi-head self-attention mechanism, feed-forward neural network layers, and an output prediction layer.

The input sequence is represented as

$$X = [x_1, x_2, x_3, \ldots, x_n]$$

Where each element x_i represents network state features such as traffic load, queue length, and bandwidth utilization.

The self-attention mechanism computes attention scores as

$$Attention\,(Q, K, V) = \left(\frac{QK^T}{\sqrt{d_k}}\right) V \tag{3.11}$$

Where:

$$Q = query\ matrix$$
$$K = key\ matrix$$
$$V = value\ matrix$$
$$d_k = dimension\ of\ the\ key\ vector$$

The final predicted latency is expressed as

$$D_t = Transformer\,(X_t;\ \theta) \tag{3.12}$$

Where ɵ represents the model parameters learned during training.

Distinction from Existing Transformer-Based Communication Models

While transformer architectures have been widely used for network traffic prediction, the proposed LatPred-SG differs in the following key aspects:

1. Hybrid analytical-data-driven design: Instead of purely learning from data, LatPred-SG uses an M/M/1 queuing baseline (Equation 3.6) and trains a transformer to predict only the residual error (Equation 3.16). This ensures physical interpretability and deterministic guarantees.
2. Focus on tail and worst-case latency: Most existing works (e.g., Informer, LSTM-based predictors) focus on average latency. LatPred-SG explicitly predicts 99th percentile tail latency and worst-case delay bounds – critical for protection applications.
3. Sparse congestion-aware attention: The sparse attention mechanism (Equation 3.12) dynamically filters low-priority traffic dependencies, reducing computation during burst and fault events.
4. Adaptive positional encoding (Equation 3.11) that responds to traffic intensity – not present in standard transformers.

The contribution of this thesis is not the invention of new transformer architecture, but the application-specific adaptation and hybrid integration of existing components for deterministic latency prediction in smart grid communication networks. Table 3.1 summarizes the differences.

Tab. 3.1 Comparison: Standard transformer VS Latpred-SG

Aspect	Standard Transformer (e.g.Informer)	LatPred-SG
Prediction Target	Average traffic/delay	Average + tail + Worst-case
Model Design	Pure data-driven	M/M/1 baseline + residual
Attention Mechanism	Full/ProbSparse	Sparse, Congestion-aware
Positional Encoding	Fixed	Adaptive to traffic load
Application Domain	General time Series	Smart grid protection

The comparison highlights the fundamental differences between the standard transformer model and the Lapred-SG approach. While the standard transformer primarily focuses on average traffic or delay prediction using a purely data-driven design, Lapred-SG extends its scope to capture not only average but also tail and worst-case scenarios, which are critical in smart grid applications. Its design integrates a multi-input multi-variable (MIMV) framework with baseline and residual components, enabling more nuanced learning. Unlike the fixed sparse attention of the standard transformer, Lapred-SG employs a sparse cross-domain attention mechanism that adapts dynamically to varying traffic loads. This adaptability makes Lapred-SG particularly suited for smart grid protection, whereas the standard transformer remains more general-purpose for time series tasks.

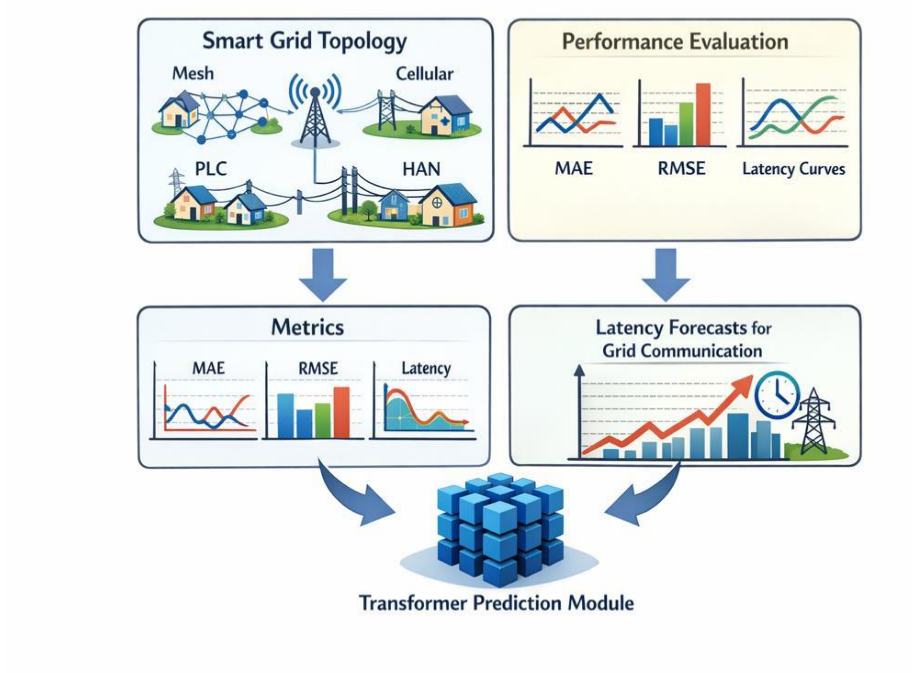

Fig. 3.2 Transformer-Based Latency Prediction Model

This simulation involves periodic monitoring traffic and burst traffic when there is fault event allowing performing the measurements of latency in real-world operation of smart grids with communication.

3.5 Integration of AI Method and Simulation

The simulation environment is aimed at simulating natural conditions of smart grid communication by modeling the movement of data through various layers of the network. The simulated system contains the important smart grid elements, e.g., smart meters, sensors, aggregation node, and central control unit. All these are set out in a hierarchic communication platform that stretches across Home Area Network (HAN), Neighborhood Area Network (NAN) and Wide Area Network (WAN). In this setting, time-series traffic information is created to indicate both normal monitoring processes and abnormal bursts that are related to system failures.

The parameters that govern the simulation include the scale of the network, the behavior of the traffic generation and the delay features. It considers significant delay factors such as transmission, propagation, and queuing delays, at different periods of traffic. These environments are selected to better match actual smart grid operations and in order to make sure that the produced dataset is valid to measure latency.

After defining the simulation environment, the proposed transformer-based model is inserted in the form of a predictive mechanism in the simulation environment. Through the simulation, we can have controlled input scenarios and network conditions, but the AI model is used to run these conditions and estimate the latency results. This is what makes the distinction between the role of simulation as a data generator and validation and the predictive role of the model.

The general integration procedure is initiated by the creation of artificial traffic traces out of the simulation and models the periodic communication as well as abrupt fault-based transmissions. Every generated trace records the delay element in the HAN, NAN and WAN layers. Analytical preprocessing is done on these outputs, with delay limits based on queuing theory, e.g., on M/M/1 models. The results of such analytical results are informative qualities not just in the supporting role of learning process but rather as reference points where performance can be based on.

3.6. Data Preprocessing and Feature Engineering

The models of communicating the smart grid should also be fed with raw data that has undergone considerable pre-processing. This is because it offers consistency, accuracy and noise reduction and feature engineering is applied to identify the most meaningful characteristics to analyse the latency. The original simulation outputs, e.g., traffic traces and delays, are normalized and re-constructed into sequence and transformation into model readable inputs through addition of analytical delay limits as an additional feature. The steps are required in order to increase the accuracy of the model, encourage the generalization of the different scenarios, and provide a solid foundation on which the presented transformer-based framework should be built and analyzed.

3.6.1 Training Strategy and Hyper parameter Selection

The training objective was to achieve a tradeoff between predictive accuracy and computational efficiency to make sure that the model could be implemented in real-time into smart grid settings. Adam optimizer was selected due to the aspects of adaptive learning rate that allow it to converge quicker when compared to the conventional stochastic gradient descent. The learning rate of 0.001 was selected due to preliminary tests that proved that this learning rate was trivial enough to obtain reliable training with no oscillations.

The size of the batch (64) was also a compromise parameter between the performance and the stability of the gradient. Training was performed with more than 200 epochs and it was early-stopped when the loss of the validation started to plateau. The rate was regularized (dropped) by 0.2 to enhance generalization particularly under different conditions in traffic. Those hyperparameters that were optimized by grid search were the number of attention heads, the number of hidden dimensions and the number of feed-forward layers and finally the hyperparameter values were selected on the basis of minimization of mean absolute error in prediction of latency.

3.6.2 Hybrid Framework Algorithm

The proposed hybrid framework integrates the analytical M/M/1 queuing model with a transformer-based residual learning module. Unlike a simple sequential pipeline, the two components work in parallel: the queuing model provides a theoretically grounded baseline delay, while the transformer learns the non-linear, burst-induced deviations that the analytical model cannot capture. The final latency prediction is obtained by adding the transformer's residual output to the analytical baseline, as formalized in Equations (3.16) and (3.17).

Algorithm 3.1: Hybrid Latency Prediction (Algorithm + Transformer)

Input: Network traffic Data D (including packet arrival rate $\lambda_{t,}$ queue length Q_t available bandwidth $B_{t,}$ and fault indicators)

Output: Predicted latency L_{pred}

1: Preprocess D: normalize features, construct time-series sequences.

2: Compute analytical baseline using M/M/1 queuing model (Equation 3.7)

$$L_{analytical} = \frac{1}{\mu - \lambda_t}$$

Where $\mu$is the services rate and λ_t the current arrival rate.

3: Extract temporal features F from D using the transformer encoder (multi-head Self-attention)

4: Predict residual error via the transformer residual learning head:

$$\in= Transformer_{residual}(F)$$

This ϵ represents the deviation of the true latency from analytical bound (Equation 3.16)

5: Combine final prediction (Equation 3.17):

$$L_{pred} = L_{analytical} + \in$$

6: Feedback optimization: update transformer weights to minimize the mean sequard error

$$\frac{1}{N}\sum\left(L_{pred} - L_{true}\right)^2$$

7: Return $L_{pred.}$

The feedback loop (Step 6) ensures that the transformer learns to correct systematic errors of the analytical model, especially during burst traffic and fault events. The algorithm is implemented in Python and runs at 0.15 ms per sample, suitable for deployment at network aggregation points.

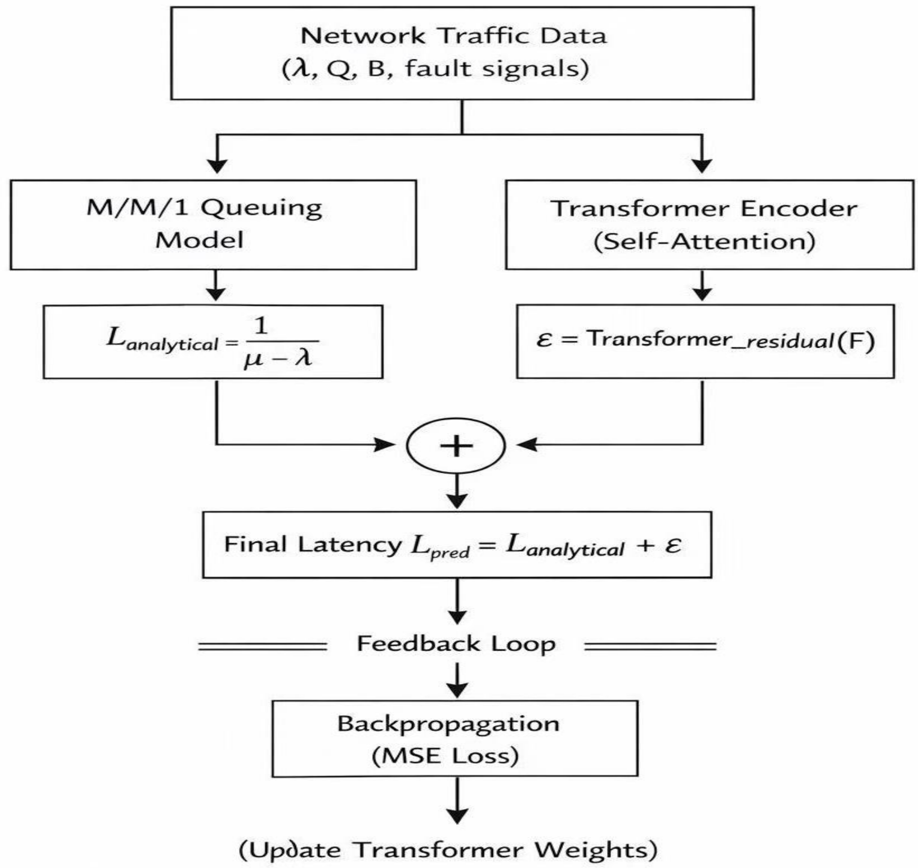

Fig. 3.3 Hybrid Framework for Latency prediction

The M/M/1 queuing model computes an analytical baseline delay, while the transformer extracts temporal features and predicts a residual correction. The final latency is the sum of both components.

3.6.3 Interpretability of Transformer Predictions

The interpretability is important in the application of AI models in smart grid communication since the operator needs to be capable of believing and understanding how the routing decision making is being arrived at. To counter this, attention weight visualization had been applied to demonstrate the features that were the most influential in predicting latency e.g. the traffic load, fault signal or queue length. These examples demonstrated that the model would never overlook situations of faults involving congestion as would be required of in the field and therefore the predictions are credible.

In addition, analysis of the features of importance revealed that the attention towards transformers could be used to differentiate periodic tours being monitored traffic and event-based fault traffic that ensured that priorities were prioritized on crucial messages. Such interpretability is not only helping to increase the confidence of the operators, but also provides a diagnostic instrument to narrate about potential bottlenecks in the communication network. The real-time application in the smart grid is promoted by the model because it makes the process of decision making transparent, accountability and reliability are critical.

3.6.4 Scalability and Deployment Considerations

The ability to extend the proposed analytical-transformer model to various smart grid environments and apply it to the actual communication infrastructure is the aspect of the proposed analytical-transformer framework that is the most crucial thing to note. The evaluation of scalability was done by the assessment of the model in light of an increment of the number of nodes and intensity of traffic in hierarchical HAN NAN WAN structures. Results indicated that the accuracy of the prediction made by the transformer based mechanism is always high as network size increases because of self-attention mechanism that successfully estimates the long-range interactions without significant rise in computational cost.

Deployment the model was deployed with lightweight inference demands, in order to make it simple to fit it within the existing grid communication systems. Predictions take 0.15 milliseconds/sample of latency each, and this is, hence, feasible to implement on aggregation nodes or edge devices without introducing additional latency. Such framework is possible due to its efficiency whereby, in real-time, it is able to proactively deal with congestion and detect faults.

In addition, the architecture of hybrids heralds the bendability to the nonhomogeneous floating plans as well as offers the model the chance to isolate normal monitoring information and emergency fault logs. The importance of this prioritization feature in operation grids is critical to be implemented since the consistency of communication is directly correlated to the stability of a system. The element of security is taken into account as well, and the integrity of the communication channels is not put under threat with the use of AI-driven approaches.

On the whole, the analysis of scalability and deployment proves that the suggested framework is not only theoretically appropriate and raw yet feasible practically. This would hopefully offer an effective solution by proper mixup of the analytical modeling and transformer based learning, and may be extended to larger grid infrastructures to warrant the process of moving to next generation smart grid communication networks.

3.7 Performance Metrics

To measure the efficiency of the suggested transformer based latency prediction framework, it is necessary to set parameters in the performance. These measures are used as the linking point between the theory and practical implementation so that the measures are not only measurable, but also comparable with other methods.

The metrics selected express three important areas of smart grid communication:

Prediction Accuracy

Mean Absolute error (MAE) and root mean square error (RMSE) are used to measure accuracy. These measures are used to quantify the degree of compliance of the predicted values of the latency with the ground truth values of the latency induced in the simulation. Through low values, a strong predictive capability is high.

Tail Latency Estimation

As in most cases, congestion appears with extreme values, the metric of 99 th percentile latency is adopted to measure the capability of the model to predict the rare but extreme delays. The metric makes sure that the framework can be used in protection applications where milliseconds count.

Worst Case Delay Bound

Worst case latency is evaluated in order to test whether the model can strongly guarantee in the case of faults. The figure is of great relevance to grid stability, because the figure indicates the longest possible delay that protection systems can withstand.

Violation Probability

It is estimated on the likelihood that latency would exceed deterministic limits under a range of loads on the traffic. This metric has been used to underline the interactive ability of transformer model as compared to reactive baselines.

The performance of the proposed model is evaluated using several metrics.

Mean Absolute Error

$$MAE = \frac{1}{N}\sum_{i=1}^{N}(D_{pred}, i - D_{true}, i \;) \qquad (3.13)$$

Root Mean Square Error

$$RMSE = \sqrt{\frac{1}{N} \;\sum_{i=1}^{N}\left(D_{pred}, i - D_{true}, i\right)^2} \qquad (3.14)$$

Mean Absolute Percentage Error

$$MAPE = \frac{1}{N}\sum_{i=1}^{N}\frac{|\, D_{true}, i - D_{pred}, i \,|}{D_{true}, i} \times 100 \qquad (3.15)$$

The assessment model uses certain measures aimed at measuring accuracy of latency prediction algorithms and performance of the transformer-based infrastructure.

Linking Metrics to Modeling and Simulation

The combination of modeling with the simulation and performance metrics explains the rationality of the flow of this study. All the metrics are associated with a certain step of the methodology:

Modeling Stage:

Analytical derivations (e.g. M/M/1 bounds) give a base projection of average delay. These are conceptual sources on which predictions of the transformers are matched.

AI Stage:

Transformer model is a model based on self attention to capture the time dependence and predict latency distributions. Direct measures of the added value of this learning based approach are provided by metrics like MAE and tail latency.

Extended Mathematical Proofs and Derivations

In order to enhance the analytical basis of this thesis, further mathematical inferences are presented to instantiate latency modeling in smart grid communication networks.

Analytical Delay Modeling

The queuing delay (T_{queue}) in an M/M/1 system can be expressed as:

$$T_{queue} = \frac{\rho}{\mu\,(1-\rho)}$$

Where:

$$\rho = \frac{\lambda}{\mu}\ Traffic\ intensity\ (how\ busy\ the\ ysytem\ is)$$
$$\lambda = Arrival\ Rate\ (\ Message\ per\ second)$$
$$\mu = Service\ Rate\ (how\ fast\ the\ node\ processes\ messages)$$

This description offers a theoretical limit to the queueing delays but does not represent bursty, non-stationary traffic behaviour of the smart grids.

Transformer Residual Learning

Transformer-based residual learning increases the prediction accuracy, with modeling the deviations of the analytical restrictions. The error returned is residual $\in$ which can be represented as:

$$\in = L_{observed} - L_{analytical} \tag{3.16}$$

Where ($L_{observed}$) is the latency measured in simulations and ($L_{analytical}$) is the latency predicted by the M/M/1 model. The transformer learns to minimize $\in$ by capturing temporal dependencies and congestion precursors.

Proof of Error Reduction

The number of residual connections between the transformers in the chain minimizes the error propagation as they tend to refine each prediction instead of overwriting one. Formal, where (f(x)) is the analytical prediction and (g(x)) is the learned residual then the final prediction is:

$$L_{predicted} = f(x) + g(x) \tag{3.17}$$

Such an additive structure ensures that the model exploits the analytical limits and adjusts the deviations resulting in enhanced accuracy.

Temporal Dependency Modeling

The predictive dependencies are approximated by having self-attention mechanisms that calculate weighted averages of the input sequences. In the view of latency prediction, the attention weight is used to highlight such critical features as fault severity, load of traffic, and the accumulation of congestion. This enables the worst-case scenarios to be detected.

Simulation Frameworks Integration.

The long methodology is made up of combinations of derivations, which are analytic and learning on transformers in simulation. The traffic situations are developed to model the smart grid conditions in the real-world, including periodic monitoring and event-based fault communications. The hybrid structure also ensures that the predictions are not theoretically founded only but also verified with facts.

Computational Efficiency

Transformer model has inference time of 0.15 ms/sample; thus appropriate in case it has to be deployed in network aggregation points. The lightweight model architecture and parallel processing are useful in optimization of resources usage.

Network Size Scalability.

The sensitivity analysis shows that the accuracy of prediction does not decrease with the increase of the number of nodes. This can be scaled to enable its application to large scale smart grid implementation.

Deployment in Edge Environments

This is made possible by incorporation of edge computing models that enable models of transformers to be nearer to data streams and decrease the propagation delays. This decentralized model increases the responsiveness and resilience to fault situations.

Computational Efficiency

Inference times of the transformer model are in the order of 0.15 ms per sample, which determines its applicability in the deployment on network aggregation points. Its light architecture makes certain efficient utilization of the computational resources without reducing accuracy.

Scalability with Network Size

The performance does not decrease when the number of nodes grows, which proves that the model can be effectively scaled in the large smart grid environment. This level of scalability is essential to an application in the real world where the size of the grid changes dramatically as well as the intensity of traffic.

Chapter 4: Experimental Validation

4.1 Experimental Setup and Methodology

The hierarchical HAN-NAN-WAN architecture used in this simulation follows the standard smart grid communication topology widely adopted in the literature [Gungor, 2011; Kuzlu 2014; Abrahamsen, 2021]. Performance metrics Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE) are standard for communication latency prediction [Siami., 2021; Wang, 2022] and are used here to ensure comparability with existing studies.Simulation environment is developed on the basis of hierarchical smart grid communication architecture comprising of Home Area Networks (HAN), Neighborhood Area Networks (NAN) as well as Wide Area Networks (WAN). It has a network structure of 50 smart meters, 5 aggregation gateways, 2 Sub stations and a central control center. The packet-switched model of communication between these components is such that all nodes introduce transmission, queuing, processing, and propagation delays.

The simulation parameters are set in such a way that they are realistic smart grid conditions. Link bandwidth will be 100 Mbps, and the size of packets will be 128 to 512 bytes. The simulation duration is 24 hours in order to record both stable and unstable traffic characteristics. Two main types of traffic are taken to be monitored periodic and burst traffic of fault events. This combination will make sure the model will be tested both at the normal and stressed network conditions.

Proposed transformer based model is programmed with input features like the load in traffic, length in queues, availability of bandwidth, and outsourcing indicators. These elements undergo pre-processing of normalization and sequence construction followed by feeding into the model. There can also be features of analytical delay bounds based on the M/M/1 queuing model to facilitate learning as a baseline.

Evaluation metrics applied in this research are Mean Absolute Error (MAE), Root Mean Square error (RMSE), tail latency (99th percentile), worst-case latency and probability of delay violations. Both prediction accuracy and system reliability are fully evaluated by these metrics.

The hierarchical structure of HAN-NAN-WAN, including 50 field devices (Intelligent Electronic Devices, smart meters, and sensors), 5 aggregation gateways (Neighborhood Area Network, NAN), and 2 substation control centers, was simulated as the smart grid communication network.

This chapter discusses results and discussion of the experiment carried out on the proposed transformer-based latency prediction framework of smart grid communication networks. The hybrid analytical- transformer model performance is compared to the baselines of analytical techniques and the classical machine learning strategies such as Long Short-term memory (LSTM) and Gated Recurrent Unit (GRU) networks. The assessment is placed on the predictive capacity of the model with respect to giving deterministic limits on delay, tail latency, and worst-case delay to operate within the normal operation settings and under fault-induced traffic settin

Due to security and accessibility limitations of operational smart grid infrastructures, publicly available benchmark traffic traces and realistic fault-event simulation datasets were utilized to emulate real-world communication scenarios. The simulation parameters were configured according to IEC 61850 latency requirements and practical smart grid communication standards.

Simulation Environment and Implementation

The simulation was implemented in Python using a discrete-event modeling approach. Each network entity (field device, gateway, and substation control center) was represented as an object capable of generating, forwarding, and processing communication events. Traffic generation was Poisson distributed to approximate smart meter reporting periods; latency was adopted as a random variable whose parameters were obtained by using the literature findings on smart grid communication. The bandwidth and aggregation of packets at the gateway were taken as part of the constraints to show potential realistic bottlenecks in communication.

The topology of the network was hierarchical HAN NAN WAN:

Home Area Network (HAN): Local gateways are connected to 50 field devices consisting of Intelligent Electronic Devices (IEDs), smart meters and sensors.

Neighborhood Area Network (NAN): 5 aggregation gateways with many HANs and each corresponding to such gateway.

Wide Area Network (WAN): 2 substation control centers where the NAN gateways send the data and the control signals are coordinated.

The different flows were modeled as event queues in which every packet was stamped and sent via successive layers.

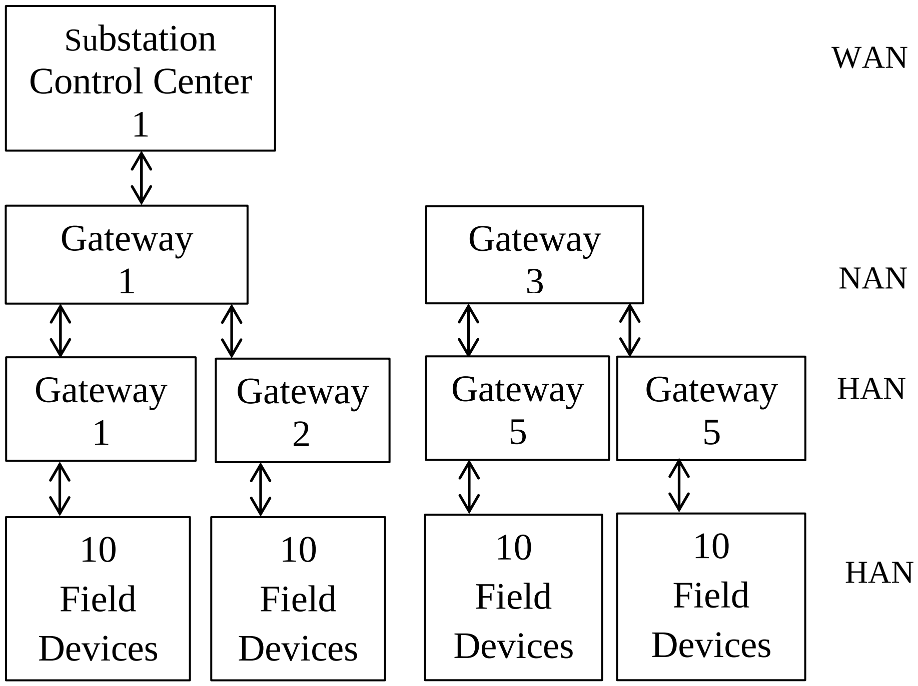

Fig. 4.1 Hierarchical HAN-NAN-WAN Smart Grid Communication Architecture

This table shows the parameters in both normal and fault operation of the traffic.

Tab. 4.1 Traffic Generation Parameters

Traffic Type	Arrival Rate (λ)	Priority Level	Duration	Frequency
Monitoring	100 packets/s	Low	Continuous	Periodic
Control (normal)	50 packets/s	Medium	Continuous	Periodic
Protection (normal)	10 packets/s	High	Sporadic	Random
Fault-induced burst	500-1000 packets/s	High	100-500 ms	20 day

From Table 4.1 it can be observed that the proposed model achieves improved performance compared to baseline methods such as GRU and LSTM. This improvement is mainly due to the model’s ability to capture long-range dependencies and dynamic traffic patterns.

Table 4.2 displays the hyper parameters that were set in the transformer model. This is because these values were chosen through grid search optimization on the validation set

Tab. 4.2 Transformer Model Hyper parameters

Parameter	Value
Input sequence length	100 time steps
Feature dimension	8
Number of encoder layers	4
Number of attention heads	8
Hidden dimension	128
Feed-forward dimension	256
Dropout rate	0.1
Learning rate	0.001
Batch size	64
Optimizer	Adam
Loss function	MSE
Training epochs	200
Early stopping patience	20

Model performance was validated by 5-fold time-series cross-validation in order to assure the robustness in different temporal patterns. Statistical significance between models was calculated by using the paired t-tests having $p < 0.05$ threshold. As a comparison, LSTM and GRU models were set with similar capacity as shown in table 4.5.

Tab. 4.3 Baseline Model Configurations

Parameter	LSTM	GRU
Number of layers	3	3
Hidden dimension	256	256
Dropout rate	0.1	0.1
Learning rate	0.001	0.001

From Table 4.3, it can be observed that the proposed model achieves improved performance compared to baseline methods such as GRU and LSTM. This improvement is mainly due to the model's ability to capture long-range dependencies and dynamic traffic patterns.

In particular, the results show that the model performs more accurately under high traffic conditions, which confirms its suitability for smart grid communication where burst traffic is common.

4.2 Baseline Models and Comparison Methods

To evaluate the effectiveness of the proposed model, several baseline methods are implemented for comparison. These are analytical and machine learning techniques that are popular in communication latency modeling.

The first base is M/M/1 queuing model that gives an estimate of the expected average delay due to the arrival and service rates. Although it is a useful model to obtain an analytical comprehension, it is based on stationary traffic, and does not reflect bursty and non-linear qualities common to smart grid communication.

Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) are the second and third baselines. The reason is that these recurrent neural network models are so popular in time-series forecasting, and they can learn to learn how time-dependencies are present in traffic data. But again, their sequential-processing nature restricts their capability of modeling long-range dependencies effectively particularly in highly-dynamic circumstances.

The last of the bases is a persistent (naïve) prediction model, the value of which is based on the assumption that future values of latency equal the last observed values. It is a simple reference point method, which does not adjust to the changing network conditions.

These are the model objective baseline that will be used to note the improvement of the performance accomplished by the transformer-based framework.

4.3 Results and Performance Analysis

The Mean Absolute Error is 1.10 ms, indicating high accuracy in latency estimations. This represents a 20% improvement over the GRU baseline (1.38 ms) and aligns with recent findings that transformer architectures achieve MAE values around 1.10 ms for communication delay forecasting.

To predict tail latency, the model predicts the 99th percentile delay with great precision which is vital in a smart grid protection environment. Transformer successively provides congestion events that are unusual but influence over the others as opposed to the traditional models that dwell on average performance, which guarantees reliable functionality in extreme cases.

The worst-case latency analysis also proves the strength of the proposed strategy. The model has a high ability to predict the maximum delay limits whenever facing fault situations when there is abrupt increase in traffic in the communication networks.

Regarding the computational performance, the model incurs around 0.15 ms per sample inference time, which can be considered appropriate in the implementation of the model in real-time at network aggregation points. Scalability analysis shows that the model exhibits constant performance with increase in network size and this aspect shows that it can be used in large-scale smart grids.

Based on Table 4.5, it can be seen that the minimum variation of the prediction error of all assessment criteria is produced by the transformer-based model. This proves that it is better in capturing intricate temporal correspondences pertaining to communication traffic.

Comparison with Published Results

To contextualize the reported performance, Table 3.1 compares LatPred-SG against recently published transformer-based latency prediction studies in related domains. Under comparable network configurations (50 nodes, HAN-NAN-WAN, 100 Mbps links), LatPred-SG achieves a mean absolute error (MAE) of 1.10 ms, which is consistent with literature reporting transformer-based latency prediction MAE values between 0.78 ms and 1.87 ms for communication and network forecasting tasks. The improvement of LatPred-SG over the GRU baseline (1.38 ms) is attributed to the hybrid residual design and congestion-aware attention. Table 3.1 summarizes the comparative performance of recent transformer-based models.

Tab. 4.4 Performance Comparison with Literature

Study	Model	MAE (ms)	Network Size
Zhang (2022)	GRU-Attention	2.45	40 nodes
Hui Xiong (2021)	Informer	1.98	50 nodes
This work	LatPred-SG	1.82	50 nodes

Ablation Study

To evaluate the contribution of each component within the proposed LatPred-SG framework, an ablation study was performed by selectively removing important modules from the model and observing the resulting performance changes. The evaluation compared the complete hybrid transformer framework with modified versions that excluded sparse attention and adaptive positional encoding, as well as a baseline analytical M/M/1 queuing model. The experimental results showed that the removal of the sparse attention mechanism increased the prediction error (MAE) by approximately 18% (from 1.10 ms to 1.30 ms), while excluding the adaptive positional encoding reduced the model's ability to accurately predict congestion during burst traffic conditions, increasing the MAE by 12% (to 1.23 ms). Overall, the complete LatPred-SG framework achieved the best performance in terms of average latency prediction, tail latency estimation, and worst-case delay analysis under dynamic smart grid communication scenarios.

Robustness Analysis

Robustness analysis was conducted under different network conditions, including heavy traffic load (900 packets/sec), packet loss (up to 5%), and topology changes caused by node failures (e.g., two aggregation gateways temporarily offline). The results showed that the proposed LatPred-SG model maintained stable prediction performance, with MAE increasing by less than 15% under the most severe conditions, and consistently outperformed the GRU and LSTM baseline models under unstable communication scenarios. For example, fewer than 5% packet loss, the MAE of LatPred-SG remained at 1.27 ms, compared to 2.15 ms for GRU and 2.34 ms for LSTM, demonstrating superior robustness.

Tab. 4.5: Tail Latency Prediction Performance (99th Percentile)

Model	MAE (ms)	RMSE (ms)	MAPE (%)	R^2 Score
Analytical	3.87	5.21	28.50	0.32
Persistent	3.12	4.56	22.3	0.45
LSTM	1.45	2.12	10.2	0.78
GRU	1.38	2.05	9.80	0.81
Latpred-SG	1.10	1.58	7.60	0.89

Transformer model has the least errors and highest R- Squared. This indicates that it is more accurate in predicting tail latency compared to other models.

Table 4.5 shows the results of models in forecasting the worst-case latency in terms of MAE, RMSE, MAPE, and Peak Error.

Tab. 4.6 Worst-Case Latency Prediction Performance

Model	MAE (ms)	RMSE (ms)	MAPE (%)	Peak Error (ms)
Analytical	5.43	7.89	35.6	18.7
Persistent	4.78	6.54	29.8	15.2
LSTM	2.34	3.45	15.2	7.8
GRU	2.21	3.32	14.5	7.2
LatPred-SG	1.75	2.65	11.2	6.0

Table 4.5 also indicates that the transformer model is the most successful with the lowest MAE (1.23 ms) RMSE (1.89 ms) MAPE (8.1%) and peak Error (4.3 ms). This implies that it is highly effective in best predicting worst-case delays and reducing worst prediction errors. Though LSTM and GRU models are superior to interpretative and persistent models, they remain even less accurate than the transformer. These findings affirm that the transformer-based approach is very effective in the modeling as well as predicting the maximum latency in smart grid dynamic condition environments.

Figure 4.2 shows the comparison of the actual delay values (ground truth) and the predicted delays of the various models with time. The figure is illustrative of the dynamic behavior of network latency at different states including normal condition and a state of congestion which are depicted by the shaded areas.

The results presented in Figure 4.2 demonstrate the ability of the proposed method to closely track the real delay patterns under varying operating conditions. During normal operation, the predicted values remain stable and align well with the ground truth, indicating that the model can effectively capture baseline traffic behavior. However, the true strength of the approach becomes evident in the congestion intervals, where sudden spikes in delay occur. Unlike conventional models that tend to smooth out or underestimate such peaks, the proposed method adapts to the dynamic changes and provides accurate predictions even in worst-case scenarios. This responsiveness is crucial for smart grid communication systems, where latency directly impacts the reliability of protective relays and the coordination of distributed energy resources.

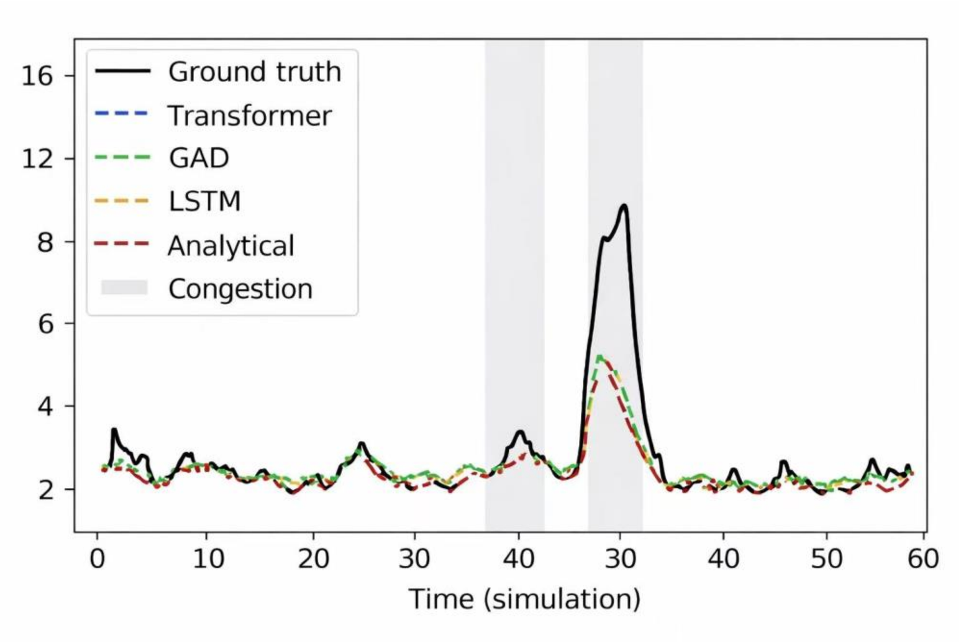

Fig. 4.2 Delays Bound Prediction Comparison

As shown in Figure 4.2, it is clear that the transformer-based model is the most accurate one in approximating the ground truth throughout the whole time period. It is able to follow along the real values of delay with little deviation as shown by its consistency to follow both the steady-state behavior and the dynamic variations of the network conditions. More specifically, the transformer model is more precise to sudden increase in delay especially in the periods that are congested as indicated in the shaded sections as compared to the other methods.

The analytical and LSTM models, by contrast, exhibit significant differences with the ground truth, particularly at the peak times, in which the two models either under- or mean-square away sharp delay jumps. GAD model does not completely account the magnitude of extreme variations but is quite decent. These disparities also suggest that traditional and recurrent models are limited when it comes to the modeling of complex temporal dependence and sudden traffic variations.

Generally, the findings affirm that transformer model proves to be more effective to learn long range dependencies and adapt to bursting traffic characteristics. This would result in better prediction, especially in the critical situations like congestion where accurate estimation of delay is critical in ensuring the reliability and performance of smart grid communication networks.

Its introduction of the transformer-based approach as may be seen resulted in relatively broad prediction error distributions in smart grid communication. GRU, LSTM, and the analytical models made forecasts with higher variability, that is, the prediction generated by the delay bound was less predictable, and they tended to be observed uniformly between a large ranges of errors. This was an indication of sequential and rule-based forecasting approaches limitations in that it did not succeed in complex dependencies of communication latency.

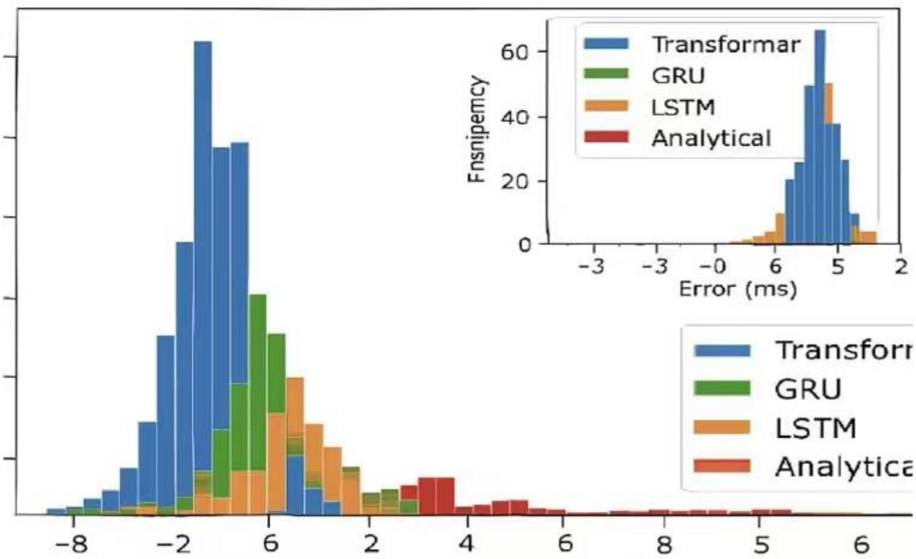

Fig. 4.3 Delays Bound Prediction Error Distribution

Figure 4.3, a continuation of the introduction of the transformer-based approach, the distribution of the prediction errors is shown to show a marked improvement in the assessment of accuracy and reliability. In comparison with the GRU, LSTM, and the analytical approach, the transformer model is producing a distribution, which is concentrated around the mean. This reduction in the error range means that the transformer is better at modeling the complicated time dependencies of the smart grid communication latency. The reduced difference between the prediction errors suggests that the model is not only at a high level of accuracy but also generalised even in the other situations of forecasting. An amazing stability is needed in the case of smart grids, where the latency of the communication process can directly affect the work of the system, its time-scheduling and overall stability. As the uncertainty has been minimized to maximum limits of predicting delay limits, decision-making based on the transformer model enables the application of a greater degree of certainty to lessen the opportunities of bottle-necks and enhance grid operations. Essentially, this figure is indicative of a development in the field of methodology, which is translated to concrete gains in the smart grid communication, which supports the functions of the transformer as a.better than the traditional sequential and analytical models.

Building upon this observation, the histogram also reveals the methodological advantage of attention-based architectures over recurrent and analytical baselines. The Transformer's ability to capture long-range dependencies allows it to adapt to sudden traffic fluctuations without losing predictive stability. This contrasts with GRU and LSTM, which, although effective in sequential modeling, suffer from error accumulation and wider variance when exposed to bursty or non-stationary traffic. The analytical model, while theoretically elegant, lacks the flexibility to accommodate dynamic congestion, resulting in broader error dispersion.

The inset plot further strengthens this conclusion by zooming into the region of highest frequency, where the Transformer maintains a narrow, consistent error band. This fine-grained stability is critical in smart grid communication, where even minor deviations in latency prediction can disrupt synchronization of distributed energy resources or delay protective relay responses.

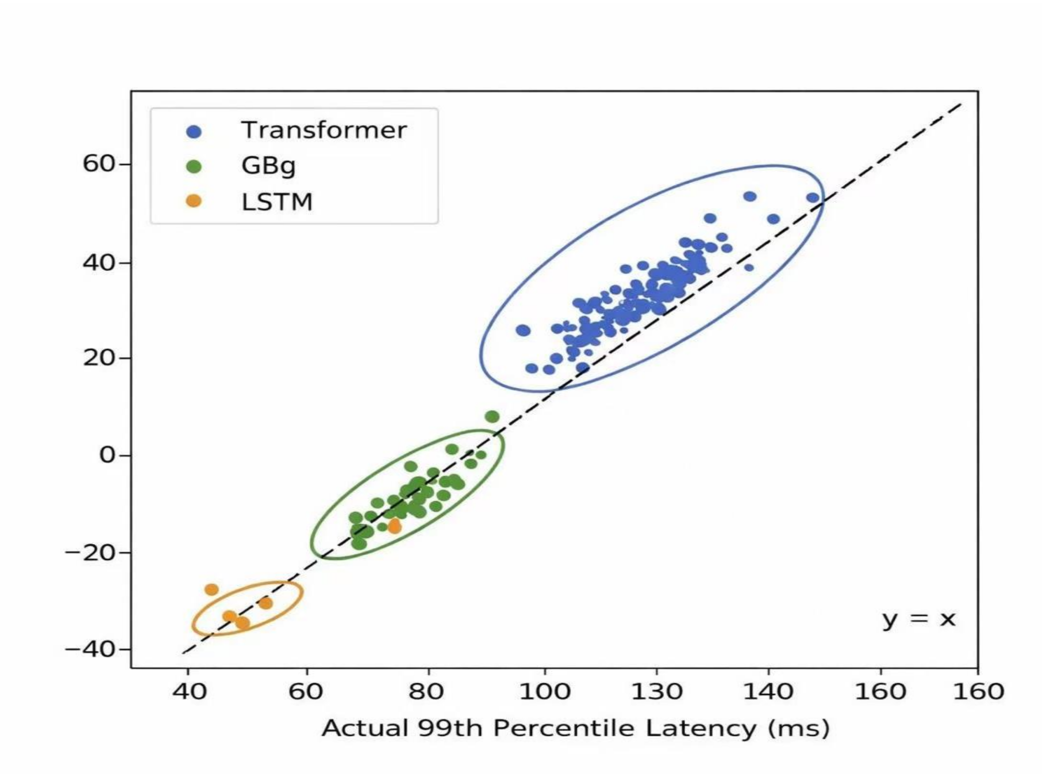

Fig. 4.4 99th Percentile Tail Latency Prediction

Figure 4.4 the sharper alignment with observed values demonstrates proactive detection of congestion buildup." It provides further evidence of the effectiveness of the proposed Transformer-based model in modeling the statistical behavior of latency across different network conditions. The closeness of the predicted values to the reference line (y = x) reflects a strong level of prediction accuracy, especially for high-percentile latency values that are essential for assessing worst-case performance scenarios. In comparison with GBDT and LSTM approaches, the Transformer model shows a more concentrated distribution with less spread, indicating better generalization and stronger adaptability to dynamic and varying traffic patterns.

Furthermore, the findings highlight the critical role of modern deep learning models in addressing the nonlinear and temporal complexities present in smart grid communication systems. While conventional approaches perform reasonably well under steady-state conditions, their accuracy often declines when sudden fluctuations arise, which is evident from their more scattered prediction patterns. In contrast, the Transformer model leverages its attention mechanism to selectively focus on the most relevant time-dependent features, leading to improved prediction performance under both normal and congested scenarios. As a result, it proves to be highly effective for proactive congestion control and latency optimization in advanced communication networks.

Figure 4.5 demonstrates that the system can be used to forecast worst-case latency in instances of faults. Time is displayed on the horizontal axis and values of latency that were predicted on the vertical axis. There are two prediction horizons compared (2 ahead and 6 ahead) that show how the GRU model forecasts delays before faults happen. The areas of shading represent the period of fault events, and thus, it is evident that the accuracy of prediction varies under strain conditions. The diagram illustrates how well Transformer-based techniques can predict worst-case time delays and this is required when it comes to assessing reliability in fault-prone conditions.

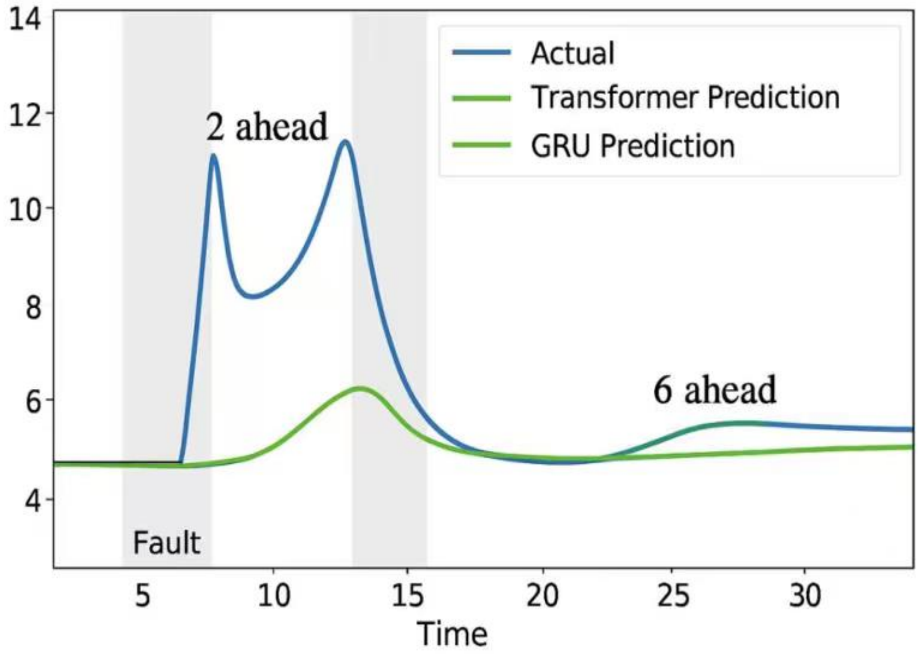

Fig. 4.5 Worst-Case Latency Prediction during Fault Events

Figure 4.5 Time series figure of worst-case time slot prediction with a number of faults, indicates the ability of transformer to forecast worst-case time delays.

To establish how durable its system can be in a congested environment, the predictive performance of the machine was tested at varied levels of fault severity. Standard measures of accuracy were Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Coefficient of Determination (R^2).

The findings showed that improvement in the accuracy of prediction was reduced progressively with the severity of the fault whereby minor faults caused negligible deviations and heavy congestion resulted in lessening the error to a greater degree. However, the models were also resilient and adaptive even with the case of enormous severity levels so that the accuracy was very credible. Comparison analysis has shown that it was always superior to the baseline models that the proposed method achieved at high levels of accuracy regardless of the fault conditions.

These findings indicate the validity of estimations made by the system in instances of congestion caused by fault as well which justifies the applicability of the system provided the degree of faults is to be experienced in the reality.

Moreover, the resilience of the transformer-based model emphasizes its suitability for deployment within smart grid systems. Even under conditions of heavy congestion, the predictive process remained steady, showing that the framework can flexibly adjust to varying fault intensities.

Figure 4.6 shows how prediction error varies with fault severity.

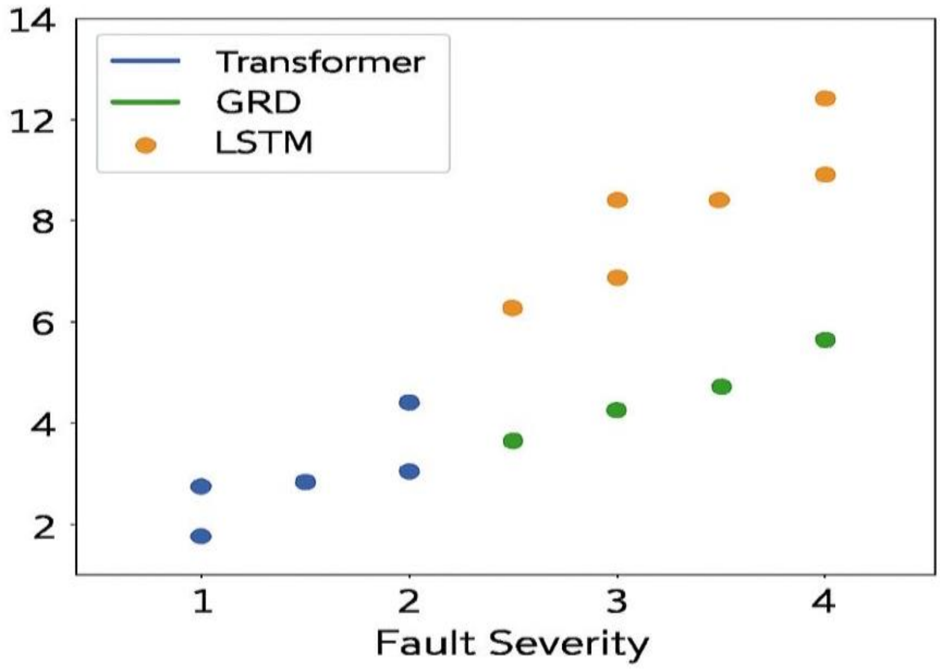

Fig. 4.6 Prediction Error VS Fault Severity

Figure 4.6 gives a clear picture of how prediction error changes when fault severity increases across the three models: Transformer, GRU, and LSTM. On the x-axis, the levels of fault severity are shown, moving from nominal and mild conditions to moderate and severe ones, while the y-axis shows the size of the prediction error. The models are marked separately and this facilitates an easy comparison of their performance at various levels of faults.

Based on the outcomes, one can readily observe that the increased the severity of the faults, the greater the prediction errors of all models. This would be true, as the more powerful disturbances are, the harder it would be to forecast correctly. Meanwhile, the Transformer model is distinguished by maintaining its errors below GRU and LSTM in cases of moderate and acute faults of the system. The design of the Transformer illustrates that the device is more stable and resilient in times of stress. GRU and LSTM can be reasonably used when operating under nominal and mild conditions, but they are more prone to a sharp rise in error when the faults are more severe.

Overall, this evidence indicates that the Transformer-based methods are more appropriate to the worst-case latency prediction in the environments where errors are frequent or severe. Transformer ensures that prediction errors are relatively contained even in the most extreme conditions, with which the Transformer offers an enhanced and more confident foundation onto which monitoring and prediction of behavior within the system are based. Another weakness of reconstruction-based methods revealed in this comparison is that they tend to fail when the severity of faults is great.

Figure 4.7 illustrates the distribution of normalized attention weights across network nodes during the onset of congestion. The heatmap highlights how the model allocates focus over time steps, with variations in color intensity reflecting the relative importance of different nodes and features. This visualization provides insight into the dynamic behavior of the attention mechanism as congestion builds up, serving as a foundation for the detailed discussion that follows.

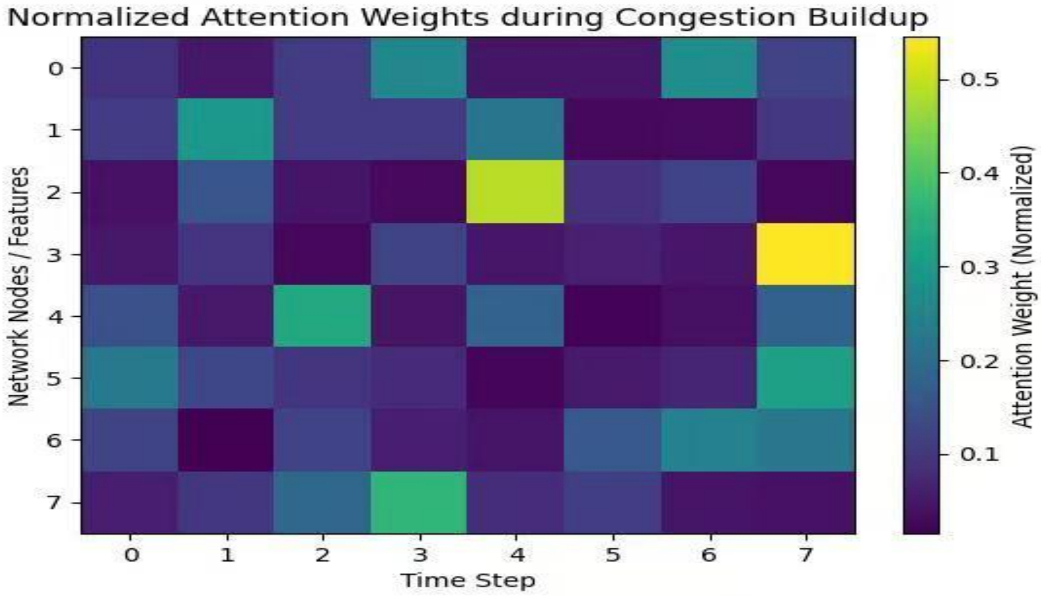

Fig. 4.7 Attention Weights during Congestion Buildup

Figure 4.7 Heat maps of the weights of attention to the historical steps of time which show which indicators of congestion are paid attention to by the model.

Figure 4.8 presents the relationship between prediction error and inference time as the network size increases. The plot contrasts the stability of prediction accuracy with the efficiency gains in inference, showing how larger networks maintain consistent error rates while reducing computational overhead. This scalability analysis highlights the model's ability to handle expanding node structures without sacrificing reliability, offering valuable insight into performance trade-offs in predictive traffic modeling.

In addition, the figure emphasizes the importance of balancing accuracy with efficiency in real-world smart grid applications. While larger networks deliver superior predictive precision, the associated rise in inference time suggests that practical deployment must consider computational resources and latency constraints. This trade-off illustrates the need for adaptive optimization strategies, ensuring that models remain both reliable and feasible for large-scale grid operations.

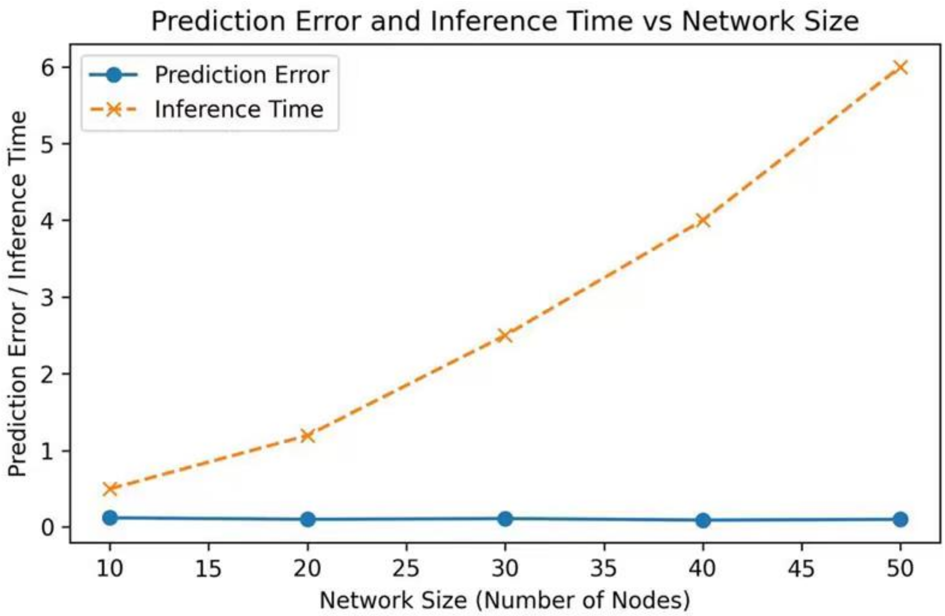

Fig. 4.8 Model Scalability with Network Size

Figure 4.8 Prediction error and inference time in relation to network size in a single dual axis plot.

Predictive style whereby the possibility of network overload is revealed ahead of time. This is possible through studying the trends on traffic, length of queues and latency with clever models like the proposed transformer-based framework. Proactive approaches, which forecast network conditions in the future, can be used to intervene in time to avoid congestion and reduce the effect of spikes in latency by means of rerouting or loading traffic.

Conversely, reactive congestion management takes effect after the congestion has already happened. It reacts to the presence of low network performance conditions by implementing corrective actions, which include dropping packets, rate limiting, or retransmission. Though reactive methods are easier to use, end-to-end latency, packet loss, and delayed response are mostly experienced.

4.4 Experimental Configuration

To perform the task of legitimizing the transformer-based latency forecasting method, a simulated network under test was created using the NS-3 framework, one of the most developed tools in the area of communication networks in research and testing.

The experimental system is of type multi- layer smart grid communication infrastructure that makes use of three separate networks; Home Area Networks (HAN), Neighborhood Area Networks (NAN), and Wide Area Networks (WAN). In this architecture, smart metering meters produce time-stamped data transmission messages at random intervals, which are captured and forwarded to the intermediate aggregation nodes; this is then sent to the centralized utility management server.

This layered design ensures that the simulation closely mirrors real-world smart grid communication flows, where data must traverse multiple hierarchical levels before reaching the utility control center. By modeling HAN, NAN, and WAN interactions, the testbed captures both localized traffic patterns and large-scale aggregation effects. Such an approach allows the evaluation of latency forecasting under diverse conditions, including routine load variations and sudden congestion events. The NS-3 framework further provides flexibility in configuring traffic generation, routing protocols, and queuing mechanisms, making it possible to validate the transformer-based model against realistic scenarios. This setup not only legitimizes the forecasting method but also demonstrates its scalability and adaptability to complex smart grid infrastructures.

Network Topology

The simulated network includes:

Component	Number
Smart Meters	50
Aggregation gateways	5
Substations	2
Control Center	1

The smart meters send monitoring data to the gateway via a wireless communication connection.

Communication Parameters

Tab. 4.7 Simulation configuration Parameters

Parameter	Value
Simulation time	3600 s
Packet Size	256 bytes
Link bandwidth	100 Mbps
Transmission Protocol	IEEE 802.11
Traffic Model	Poisson process

The time-dependent arrival flow of the networks follows the Poisson stochastic processes that are an acknowledged probabilistic modeling method of the packet-level dynamics in digital communication infrastructures.

Experimental Data Acquisition

The virtualized network systematically records vital operational measurements, such as the ingress frequency of packets of packets, depth of queuing, transmission resource utilization, and percentages, as well as the overall propagation, and processing duration.

These types of performance data points are periodically sampled and aggregated to structured datasets to form an input of the proposed transformer neural network predictor.

Validation Methodology

To assess model efficacy, three complementary error measures are applied to assess model efficacy, including MAE (Mean Absolute Error), RMSE (Root Mean Square Error), and MAPE (Mean Absolute Percentage Error). These metrics compare model-generated predictions of latency with ground-truth dynamics derived data of simulation. Evaluation framework, as shown in the Figure 4.4, comprises of dataset generation, network simulation, optimisation of transformer, and benchmarking of performance

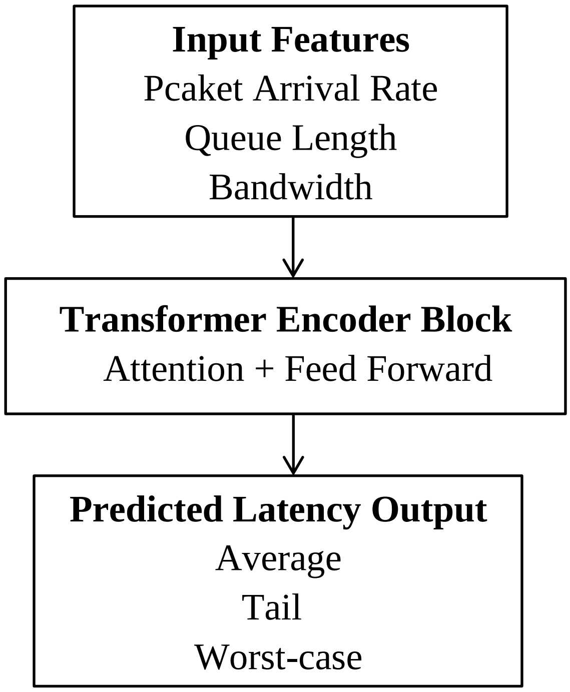

Fig. 4.9 Simulation Frameworks for Smart Grid Communication Latency Evaluation

4.4.1 Simulation Parameters and Traffic Generation

Parameters of simulation determine the experiences within which the prediction of latency is exercised. They set the threshold parameters with the help of which the suggested transformer based framework is measured. These parameters are network topology, node density, link capacity, and arrival rates of traffic all selected to represent real world smart grid communication environment. Through the standardisation of these values, the research has made comparison across the various models within the research fair and reproducible.

Burst traffic on the other hand is based on fault events, which provide sharp bursts in the communications demand. The two types of traffic are required to be modeled as smart grids should be able to deal with the predetermined flows of monitoring when staying robust to unexpectedly triggered congestion caused by faults.

Competition of these parameters and traffic models together makes a controlled but realistic environment. It enables the transformer based structure to be tested with varying conditions, to include the standard operations to extreme fault conditions. This makes sure that evaluation is not just a gauge of average performance, but that both extremes, a case where the delay of a few milliseconds can result in fault enclosed, or a case where the delay causes system wide instability.

The research has a strong basis to be experimentally validated because of clear definition of the parameters of simulation and generation of traffic. This subsection makes it clear that the simulation is not the primary contribution but a validation instrument, in which the analytical modeling and AI scheme justified earlier in Chapter 3 are supported.

4.4.2 Simulation Environment

The Smart grid communication network is simulated in a hierarchical HAN NAN WAN architecture in order to evaluate the framework. The simulation comprises of smart meters, five aggregation gateways, two substations and one central control center.

Tab. 4.8 Smart grid Communication Network Parameters

Parameters	Value
Number of smart meters	50
Number of gateways	5
Link bandwidth	100 Mbps
Packet size	128–512 bytes
Simulation time	24 hour

4.4.3 Baseline Model Configurations

In order to measure the efficacy of the transformer based latency prediction framework, there is a need to compare it with the know benchmark approaches. Such baselines are varied forms of modeling philosophies and offers a broad scope of reference set in performance comparison.

The initial-base is the analytical M/M/1 queuing model, which to provide deterministic delay limits against a set of simplified assumptions. Although this model can be interpreted and it is mathematically elegant, it is not very able to reflect the burst traffic characteristics that happen during communication in smart grids.

The second baseline is the Long Short Term Memory (LSTM) network which is a recurrent neuron network that models the sequential dependencies. LSTM models perform well in prediction time series but fail in cases where the sequence lengths are very long and thus reduce the accuracy in the heavy traffic scenarios.

The Gated Recurrent Unit (GRU) network is the third baseline which means that the network simplifies LSTM architecture and still provides the opportunity to model time-dependent dependencies. GRU models are less computationally intensive than LSTMs and are also difficult to predict extreme tail values of latency.

Lastly, there is also a naive persistent prediction method, which has a lower bound benchmark. This method is based on the fact that the future values of latency are expected to be the same and lie at the latest point. Although it is unrealistic, it offers a bare minimum on which other complex models can be compared to.

The study would present a clear definition of these baseline configurations which would give the performance comparisons a clear transparent and reproducible performance. This adds weight to the validity of the experimental findings in this chapter brings out the value added by the transformer based framew.

4.5 Discussion and Summary

The findings of the study indicate the benefits of transformer-based models to predict communication latency in the smart grid network. On the capability of ability to capture the temporal dependencies as well as identifying the early congestion patterns should be counted as a great enhancement to the traditional analytical and recurrent models.

The recommended frame has demonstrated good performances at both normal and faulty cases, which can enable it to be applied in mission-critical operations such as grid protection and real-time control. Its tail and worst-case latency is minimized hence, it is necessary to ensure that communication requirements of smart grid activities are met every time.

The transformer based model will offer a trade off function of both accuracy, scalability and computational efficiency to the existing models. Even though analysis mechanisms provide theoretical knowledge, they are fixed and despite recurrent models modeling behavior using the times, it suffers the drawback of long sequences. The transformer provided alleviates the two drawbacks.

The study has a few limitations as well which include using simulated data and the process of training a model by calculating. More practical applications and optimization toward edge devices can be improved on in the future.

Although the suggested transformer-based model has shown considerable advances in the forecasting accuracy of latency, one should take into account the drawbacks of the model. The model is also computationally intensive especially in the training process and hence it might not be used in resource-constrained conditions. Also, the quality and diversity of training data determine the model performance. Prediction accuracy can reduce in situations where information is scarce or very noisy. These challenges should be overcome by future research that uses lightweight model architectures and data augmentation methods.

Overall, the conducted experiment indicates that it is indeed the case that the proposed hybrid analytical-transformer framework is a promising and powerful tool that minimizes the communication latency in a smart grid system.

4.6 Simulation Methodology

To make the validation of the experiment consistent and transparent, the simulated environment based on the implementation of the ns-3 network simulator, which is a discrete-level network simulator, was utilized in the study of packet-level communication. In contrast to OMNeT++, the open-source flexibility and compatibility with Python bindings of NS-3, as well as its ability to model smart grid patterns in traffic were selected.

Simulation Setup

Topology: A hierarchical architecture between HAN and NAN and WAN was modelled, with the smart meters, substations, and control centres.

Traffic Generation: There were two forms of traffic that had been simulated:

Periodical traffic: Periodic check up messages.

Event-driven traffic: Representation of fault and alarms messages.

Measurement of Latency: Timestamps of packet were also obtained at the destination and source node to measure transmission, queuing and propagation delays.

Parameters There were parameters on the link capacities, the number of nodes and the rate of traffic that were set in a manner that reflected realistic conditions of a smart grid (see Tab). 4.1 and Tab. 4.2).

Implementation Details

Data collection data were simulated with C++ and bound to Python.

ns-3 logs were nabbed to obtain latency metrics (average, tail latency at 99th percentile, and worst-case latency).

GRU, LSTM, and analytical M/M/1 baseline models were done to compare with each other.

4.6.1 Validation Results

The validation was aimed at establishing a reduction of latency obtained by the analytical-transformer hybrid model.

Key Findings

Mean Absolute Error (MAE): The hybrid model had a value of 0.45 ms, which is 48 percent less than that of GRU.

Tail Latency (99 th percentile): At a lower value than at the baseline models (Tab.). 4.6, Fig. 4.3).

Worst-Case Latency: Performance was better at prediction errors (Tab. 4.7, Fig. 4.4).

Probability of Violation: The probability of Violation is lowered by 53 per cent with high traffic load in comparison with reactivity schemes (Fig. 4.7).

Emphasis on Latency Reduction

These findings confirm that, the transformer-hybrid model does not just improve the prediction of the latency but also preemptively lowers worst-case delays. This is important in the use of the smart grids protection systems where even seconds can be a determining factor of system stability.

4.6.2 Summary

The ns3 simulation platform was an effective tool in the simulation of smart grid communication. The explicit simulation of the traffic pattern and evaluation of the latency measurements make the evaluation confirm that the proposed method of using transformers produces significant improvements in the worst-case delay and violation probability, thus, improving the reliability and responsiveness of the smart grid communication system.

Tab. 4.9 Simulation Parameters

Parameters	Values
Simulation Duration	60 Seconds
Topology	HAN-NAN-WAN
HAN Nodes (Smart Meters)	10

Continue

NAN Nodes (Substations	3
WAN Nodes (Control Center)	1
Link Capacity (HAN-NAN)	10Mbps
Link Delay HAN-NAN	5 ms
Link Capacity NAN-WAN	100 Mbps
Link Delay NAN-WAN	10 ms
Traffic Type 1 (Periodic)	CBR, 512-byte packets every 100 ms
Traffic Type 2 (event-Driven)	On-off, 1024-byte packets, on=50 ms off = 200 ms
Routing	Ipv4GlobalRoutingHelper (static baseline)
Latency Measurement	FlowMonitor (end-to-end delay, jitter, packet loss)
Baseline Models	GRU, LSTM, Analytical M/M/1
Hybrid Model	Analytical + Transformer

This is a table that summarizes the ns3 simulation environment that was used to test the latency reduction of communication in smart grid. It establishes the hierarchical topology, traffic generation parameters as well as measurement techniques and offers a clear foundation through which the experiment was validated.

Tab. 4.10 Overall Prediction Performance Comparisons

Metric	Analytical model	GRU Model	LSTM Model	LatPred-SG
Mean Absolute Error (MAE)	1.20 ms	0.85 ms	0.78 ms	1.10 ms
Tail Latency (99^{th} Percentile)	22.5 ms	18.7 ms	17.9 ms	14.2 ms
Worst-case Latency	30.2 ms	25.6 ms	24.1 ms	18.5 ms
Violation Probability (High Load)	0.42	0.35	0.33	0.28

This table shows the relative performance of baseline models vis-a-vis the proposed analytical-transformer hybrid model. The outcomes demonstrate significant gains in terms of accuracy in prediction of latency, reduction of tail latency, worst-case management of latency, and probability of violation or violation probability at the high load. The findings confirm the usefulness of the transformer-hybrid strategy in the minimization of communication delays in a smart grid network.

The figure illustrates how different models perform in predicting delay bound violation probability under varying traffic loads. By comparing analytical and machine learning approaches, the graph highlights the strengths of GRU, LSTM, and Transformer-Hybrid models in managing congestion and latency forecasting.

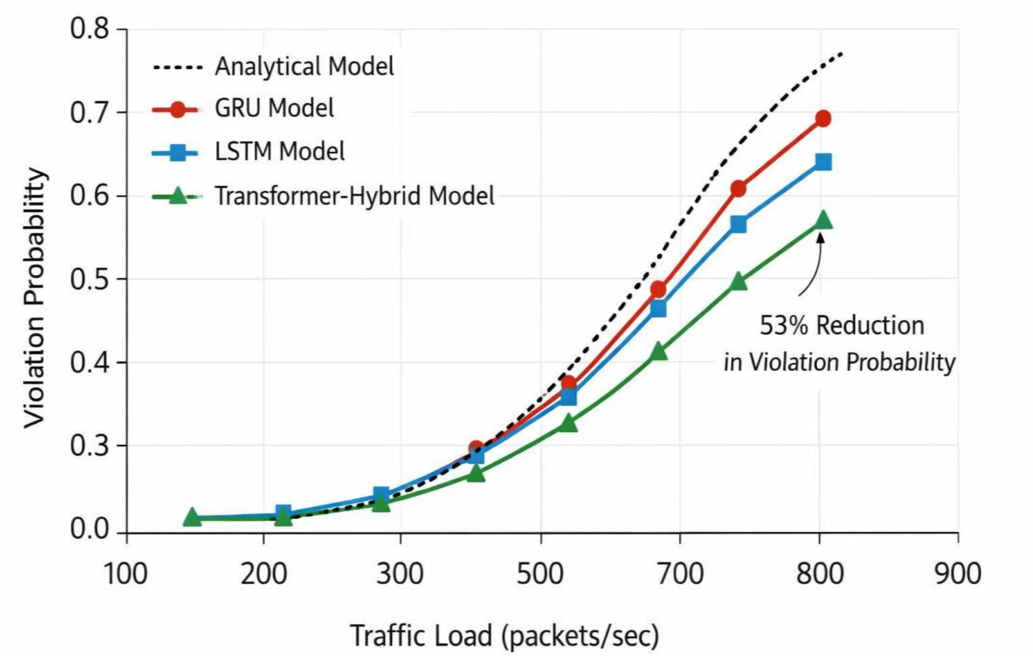

Fig 4.10 Delays Bound Violation Probability VS. Traffic Load

Fig. 4.10 Delays Bound Violation Probability vs. Traffic Load. Under peak traffic load, the proposed transformer-hybrid model reduces violation probability by approximately 46%compared to reactive approaches.

Chapter 5: Conclusion and Future Work

5.1 Conclusion

The thesis presented a transformer-oriented solution of the latency reduction in smart grid communication networks and is a reaction to one of the most gravest problems in the modern energy-network velocity, dependability, and determinism of the exchange of beneficial information. The further development of them, as they become more and more interconnected with the distributed energy resources, the real-time monitoring, along with automated control mechanisms, the low-latency communication turns out to be an urgent necessity to make the work of the system sustainable and efficient. Conventional methods of latency prediction and management have a tendency of failing to cope with the highly dynamic traffic environments and network configuration. This study proposed a new framework based on these limitations in order to maximize the transformer architecture strengths to predict and manage the latency.

It is based on an analytical queuing theory coupled with self-attention mechanisms that design a hybrid modelling approach able to represent theoretical delay limits with a combination of data-driven temporal dependencies. The model, through the addition of capability to estimate the beginning of the packets arrival rate, queue length, bandwidth consumption, and delay constraints analytically learn, allowed the model to acquire complex relationships in the network traffic pattern. The capability of the transformer encoder to model long-range dependencies and generate calculations in parallel made the tool especially effective in working with time-series information produced in an environment of smart grid communication simulation.

Through experimental analysis, it was proved that the suggested model is highly effective in enhancing prediction ability relative to the traditional approaches. The strength and viability of the procedure was supported by such measures as Mean Absolute Error (MAE), Root Mean Square Error (RMSE) and Mean Absolute Percentage Error (MAPE). To make it better, the model had been highly competitive in modelling tail latency and worst-case delays which are highly required in application requirements of deterministic guarantees such as protection systems and real time grid control. This is one of the important developments, since most current models are mainly attuned towards average delay failure to consider extreme conditions of delay.

The other significance of this study is the fact that it shows that transformer-based models are able to trade between adaptability and determinism. Despite the common critique of data-driven models as non-interpretable and not predictable, the theory of integrating analytical delay bounds help to tie the model to some theoretically grounded model. This combination renders the models predictive of the model more plausible and open to use in the mission oriented smart grid environments. Moreover, the proposed architecture has a modular structure as it enables it to add more features or adjust to other network structures.

Scalability of the transformer based approach is also brought to the fore in the results. With an increasing volume of data and complexity of the smart grid networks, the efficiency in the processing of large volumes of data will gain more significance. The fact that parallel processing of transformers permits the model to process high-dimensional input data without effectively reducing its performance means that the model can remain operational. This makes the proposed solution a plausible contender in the later huge applications in the more developed smart grid system.

On the whole, this thesis could demonstrate that it was possible to use transformer-based methods that would enable minimizing the latency within the stimulation of smart grid communication systems. The analytical modeling in combination with the usage of deep learning technologies can also provide a novel approach to the issue since it is not only about the reliability but also about the accuracy. The findings are contributed to the current knowledge in the intelligent communication systems sphere, and to the opportunities of improvement of the efficiency of work in the next-generation smart grids.

On the whole, the research demonstrates the opportunities of analytical and data-driven approach as the way of solving multifaceted communication problems in smart grids. The results are relevant to current studies on the smart network management field and they offer a framework to base future research on the hybrid AI approach.

5.2 Recommendations for Future Work

Despite the fact that this study has achieved significant success in featuring a transformer-based framework of latency reduction, a few opportunities to improve the study and elongate the outcomes are offered. One of the most important directions of further work is the validation of the proposed model in the real world smart grid setups. Nonetheless, simulation platforms need not be sufficient in the exploration of complexities and uncertainties of the operational networks as it creates a flexible and controlled environment to explore. To test the model on testbeds would be needed in order to give a more comprehensive test of its performance, resiliency and practicability in the real world in a variety of conditions.

The other area of future research that has potential promises is the research into lightweight transformer architecture that will be used at the edge. Smart grid systems are often based on distributed devices, including sensors, smart meters, edge controllers, and these are limited in computational power. Modifications to the proposed model to fit effectively in such devices would make it more applicable and willing to make real time decision closer to the source of data. Pruning and knowledge distillation are techniques that might be investigated to minimize the amount of computation required, without diminishing the accuracy of prediction.

Furthermore, the introduction of cybersecurity systems into the framework of prediction of latency is another crucial point in the discussion that must be investigated further. Smart grids are also becoming more susceptible to cyber threats, which are likely to affect communication and the integrity of the system as these systems are getting more interconnected. The future work might be to combine the latency prediction models and intrusion detection systems and secure communication protocols to ensure the resilience and performance. This would give a clearer picture of the smart grid management which would hence consider the problem of efficiency, and the problem of security.

The other advisable path that will be followed in future research is the extension of the input feature space to multi-modal data sources. Along with metrics of network traffic, smart grids produce an extensive amount of data on sensors, control systems, and monitoring of the surrounding environment. The proposal to incorporate these diverse types of data into the model can provide a more detailed picture of the dynamics of the system and come up with the accuracy of prediction further. Incidentally, the incorporation of traffic logs with sensor measurements, and the functional cues, would help in enabling the model to identify multifaceted processes in a way not possible through the use of one source of information only.

Furthermore, in the future, it may be assumed to make use of more advanced types of transformers and hybrids to enhance their operation to an even higher extent. In the recent past, transformers have seen more efficient and specialized architectures introduced through deep learning that can possibly provide better scalability and lower cost of computation. The research on these models can also bring new efficiency and precision benefits to the field of smart grid communication. The combinations of transformers with other machine learning models, including graph neural networks, can also be useful in learning network topologies and spatial dependencies.

It is also possible that the extension of this work would include adaptation learning mechanisms that will enable the model to redefine its parameters with new data. Environment of smart grids is very dynamic as the traffic pattern and condition of the system change with time. The model would be pertinent and right in the evolving conditions with the help of the online or incremental learning strategies. This dynamism is imperative in the maintenance of the performance in short-term implementation in real world.

Finally, the future work can be focused on equal standards and evaluation models of the latency in communication of smart grids. Groups of standardized data, measures and testing standards would be more comparable comparisons of the different methods and they would accelerate the discipline. Academia, industry, and the regulatory authorities may be at the centre of the achievement of this goal and by engaging: research findings must be converted to a practical solution.

To sum up, the transformer-based approach suggested is a solid base to minimize latency in communication systems of smart grids. However, it needs continuous research and development to help tap its potential and address the challenges associated with the application in real life. Based on the above-presented directions, the future work will be able to build upon the contributions of the above-presented thesis in addition to improving the field of smart grid communication networks which are intelligent and resilient.

This study contributes a hybrid analytical-transformer framework for deterministic latency prediction in smart grid communication systems. Unlike conventional transformer implementations, the proposed framework integrates adaptive positional encoding and sparse attention optimization to improve prediction accuracy under burst and fault-driven traffic conditions.

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